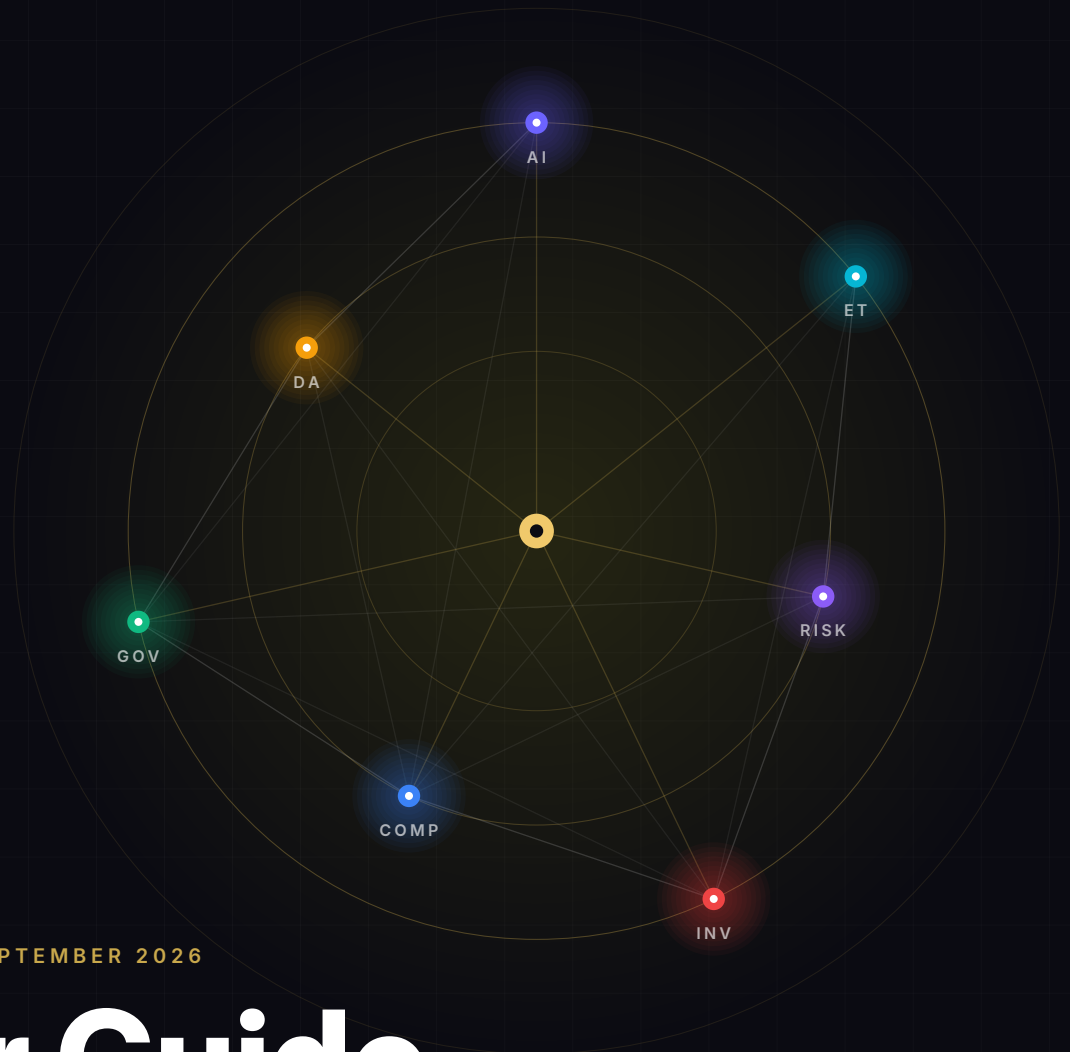




EDITION 2.0 · SEPTEMBER 2026

User Guide & Technical Review

Everything learners, partners and engineers need to know about the Chronopoli platform – illustrated end to end with a fictitious Tether (USDT) academy scenario, real screenshots and clearly labelled design prototypes.

HOSTED BY

Dubai Blockchain Center

PLATFORM

Open edX · Tutor · AWS

DISTRICTS

7 Knowledge Districts

DEMO DATA

Fictitious, for illustration

CONTENTS

What's inside

A practical guide for learners and partners, followed by an engineering review of the platform as it stands in September 2026.

PART ONE Welcome to the knowledge city

01	Chronopoli at a glance	04
02	The Tether demo scenario	06

PART TWO Learner guide

03	Getting started and AI onboarding	09
04	The Tether academy and its courses	11
05	Enrolment, pricing and payment	15
06	Learning with the AI Tutor	18
07	Grades, badges and careers	19
08	Symposia round tables	22
09	Community, video and slides	24

PART THREE Partner & admin guide

10	Running a partner academy	27
11	Partner dashboard and intelligence	29
12	Digital Twin masterclasses	31
13	Administrator quick reference	32

PART FOUR Technical review

14	Architecture overview	34
15	Third-party stack and versions	36
16	Review findings	38
17	Updates applied and road to go-live	42
18	Glossary	44

01

PART ONE

Welcome to the knowledge city

What Chronopoli is, how it is organised, and the Tether demo scenario used throughout this guide.

IN THIS PART

01	Chronopoli at a glance	04
02	The Tether demo scenario	06

— CHAPTER 01

Chronopoli at a glance

Chronopoli is a global knowledge city for the digital economy. It brings regulators, law enforcement, banks, enterprises, founders and students onto one platform. Seven Knowledge Districts organise the content, and four learning layers set how deep each learner goes.

7

KNOWLEDGE DISTRICTS
One Open edX org each

4

LEARNING LAYERS
Entry to Influence

8

CUSTOM DJANGO APPS
Not yet installed (R-1)

0

FORKS OF OPEN EDX
Stays upgradable

Seven Knowledge Districts

Each district is an Open edX organisation with its own colour, audience and course tracks. A course always belongs to one district, and a partner (in this guide, the illustrative Tether academy) can publish tracks into several districts.

**AI District · CHRON-AI**

AI governance, model risk and enterprise AI for CIOs and executive teams.

**Digital Assets · CHRON-DA**

Blockchain, tokenisation, stablecoins and custody for bankers, founders and regulators.

**Governance · CHRON-GOV**

Policy, regulation and public-private dialogue for ministries and policymakers.

**Compliance · CHRON-COMP**

AML/CFT, KYC, sanctions and the Travel Rule for MLROs and compliance teams.

**Investigation · CHRON-INV**

Blockchain forensics and financial crime cases for law enforcement and FIUs.

**Risk & Trust · CHRON-RISK**

Operational, cyber and treasury risk for CROs and risk managers.

**Emerging Tech · CHRON-ET**

Quantum, IoT and spatial computing, and what comes after today's stack.

Four learning layers

LAYER	TAG	WHO IT IS FOR	TYPICAL FORMAT	PRICE BAND
L1 Entry	layer:entry	Students, newcomers, the interested public	Short self-paced courses	\$0–\$149
L2 Professional	layer:professional	Compliance teams, executives, founders	Self-paced + verified certificate	\$299–\$799
L3 Institutional	layer:institutional	Governments, regulators, enterprise	Advanced practitioner programmes	\$999–\$2,499
L4 Influence	layer:influence	Thought leaders, researchers, policy shapers	Live cohorts, round tables	\$3,000–\$8,000

How partners take part

Organisations join as **Knowledge Partners**. They host district-branded tracks and can run a white-label **Company Academy** on their own subdomain. Partners are grouped into four tiers.

TIER	EXAMPLES FROM THE PARTNER ROADMAP	WHAT THEY GET
Founding	Ripple, Cardano, Hedera	Flagship tracks, co-hosted symposia and a seat on the curriculum board
Strategic	Tether, Polygon, Chainlink	Partner academy, pathways and badges, talent pipeline
Content	Chainalysis, TRM Labs, Elliptic	Specialist courses inside a district
Institutional	Universities, think tanks, banks	Co-certification and research round tables

● IMPORTANT

The partner names above come from the Chronopoli partner roadmap. They do not confirm any signed agreement. All Tether figures, people and courses in this booklet, and the data shown on its screens, are **fictitious mock data** that show how the platform works.

● SCOPE

Parts I to III describe how the platform is **designed to behave** at go-live. The technical review in Part IV lists the gaps found in the current code that must be closed before go-live, and the updates already applied. Where a screen or a step depends on an open finding, the finding is named (for example R-10).

— CHAPTER 02

The Tether demo scenario

This guide follows one story from start to finish: Tether opens an academy on Chronopoli, five learners use it, a partner administrator runs it, a Chronopoli moderator hosts its round table and one expert teaches through it. Every name, number and course in it is fictitious; the screens show real software, or clearly labelled prototypes, filled with this mock data.

The partner

FIELD	MOCK VALUE	WHERE IT LIVES IN THE PLATFORM
Partner name	Tether (illustrative)	<code>chronopoli_partners.Partner.name</code>
Tier	Strategic Partner	<code>Partner.tier = strategic</code>
Open edX org code	TETHER (partner identifier; Tether courses stay under the district organisations, see chapter 4)	<code>Partner.openedx_org_code</code>
Districts	CHRON-DA, CHRON-COMP, CHRON-INV, CHRON-RISK, CHRON-GOV	<code>Partner.districts (JSON)</code>
Academy	Tether Stablecoin Infrastructure Academy on <code>tether.chronopoli.io</code> ; its header reads 'Tether (illustrative) Academy'	<code>Partner.subdomain + academy_enabled</code>
Brand colours	#26A17B / #0F1F1A (stored, not yet applied)	<code>brand_primary_color / brand_secondary_color</code>
Payouts	Stripe Connect acct_1TthrDEMO0000	<code>Partner.stripe_connect_account_id</code>
HR notifications	talent@partner.example	<code>Partner.hr_notification_email</code>

● HOW TO READ THIS GUIDE

Layla is the main character: **chapters 3 to 9** follow her from sign-up to her CSCS badge, the open roles it unlocks and the community, including the round table that Dr. Haddad moderates (chapter 8). **Chapters 10 to 13** cover the operator side: Nadia runs the academy and its dashboard, and Prof. Weiss is the expert behind the Digital Twin.

How to read the figures

Every figure was made for this edition. Its caption starts with a label that says what you are looking at:

CAPTION LABEL	WHAT IT MEANS
Real Open edX screen	Captured from Open edX (Tutor 22.0.2, Verawood) running in Docker with the demo courses loaded.
Real Chronopoli template, rendered with demo data	A repository template, rendered by its real Django view in a test harness with mock data.
Real extension (Discourse / Opencast)	Captured from Discourse 3.5 or Opencast 20.4 in Docker, filled with demo content.
Design prototype	The page does not exist yet (in most cases because its template is missing, R-10) or, for the AI Tutor, the real widget is shown on a composed lesson page. Prototypes show the target design built on the real data model.
Diagram	An explanatory drawing of the target architecture, not a screen.

Browser frames and addresses were added for orientation; they show where each page will live after go-live, not the local address used for capture.

The people

Five learners, a partner administrator, a Chronopoli moderator and one expert take part. The learners' onboarding answers are fed through the real routing logic in `chronopoli_onboarding.calculate_recommendations()`; the table shows where each of them is placed.

PERSONA	ROLE (FICTITIOUS)	Q1 ROLE / Q2 FOCUS / Q3 EXPERIENCE	ROUTED TO / WORKS IN
Layla Al-Mansoori	MLRO, Gulf Digital Exchange (VASP)	exchange / CHRON-COMP / advanced	Compliance · L2
Omar Farouk	Financial investigator, national FIU	law_enforcement / CHRON-INV / intermediate	Investigation · L2
Priya Nair	Head of Treasury Risk, regional bank	bank / CHRON-RISK / expert	Risk & Trust · L2
Mariam Saleh	Policy adviser, finance ministry	regulator / CHRON-GOV / expert	Governance · L3
Jonas Weber	Master's student, fintech	student / CHRON-DA / beginner	Digital Assets · L1
Nadia Karim	Academy administrator (partner side, fictitious)	Not onboarded (partner staff)	Academy and dashboard
Dr. Amira Haddad	Round-table moderator, Chronopoli	Not onboarded (staff role)	Symposia approval
Prof. Daniel Weiss	Stablecoin economist (expert)	Not onboarded (expert account)	Digital Twin

02

PART TWO

Learner guide

From sign-up to the CSCS badge: onboarding, the illustrative Tether academy, courses, payment, the AI Tutor, careers, round tables and the community.

IN THIS PART

03	Getting started and AI onboarding	09
04	The Tether academy and its courses	11
05	Enrolment, pricing and payment	15
06	Learning with the AI Tutor	18
07	Grades, badges and careers	19
08	Symposia round tables	22
09	Community, video and slides	24

— CHAPTER 03

Getting started and AI onboarding

You create one account for learning and the community. A five-question onboarding then builds your personal map of the knowledge city: a home district, two neighbouring districts and the learning layer to start in.

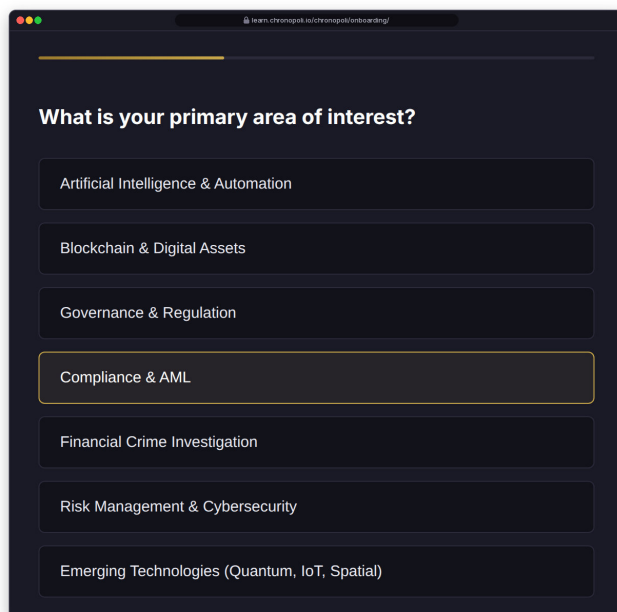
Create your account

- 1 Open learn.chronopoli.io and choose 'Sign Up' (top right)**
Use your work e-mail if your employer holds a team licence, so that an administrator can assign you a seat.
- 2 Confirm your e-mail address**
Amazon SES sends the confirmation link.
- 3 Start the onboarding**
Open /chronopoli/onboarding/ after your first sign-in (an automatic first-login redirect is planned). If you skip it, you can still browse the full catalogue.
- 4 Community sign-on**
The same account signs you in to the Community (Discourse SSO). Lecture recordings play in the Opencast video player (video.chronopoli.io, chapter 9); embedding them in course units via LTI is planned. The Slides builder is for instructors and currently has a separate staff login, with single sign-on planned for Phase 2.

The five onboarding questions

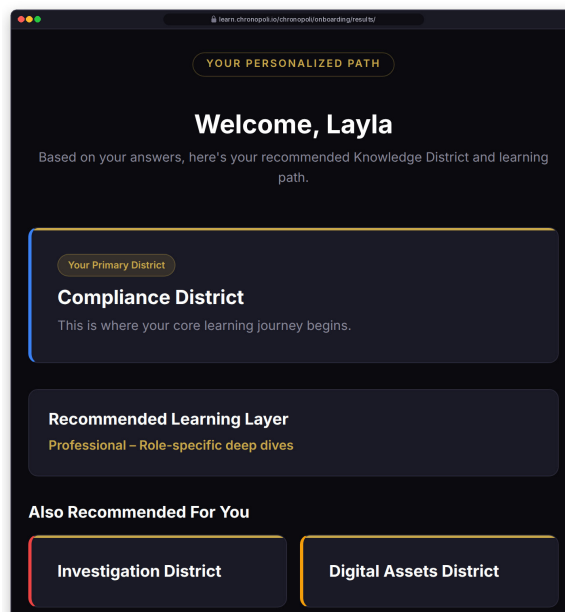
Each question is one screen with one answer to pick (labels shortened here; the screen shows the full wording).

#	QUESTION	ANSWER OPTIONS (SHORT LABELS)	LAYLA'S ANSWER
Q1	Which best describes your role?	Regulator, Law Enforcement, Bank, Crypto Exchange, Enterprise Leader, Founder, Student, Consultant	Crypto Exchange
Q2	What is your primary area of interest?	One option per district, e.g. Compliance & AML	Compliance & AML
Q3	How would you describe your experience in this field?	Just getting started, Some experience, Experienced professional, Senior expert	Experienced professional (stored as advanced)
Q4	What is your primary goal on Chronopoli?	Market awareness, Practical skills, Professional certificate, Policy dialogue, Expert network	Professional certificate
Q5	How urgent is your learning need?	Now, Within weeks, Over months, Just exploring	Within weeks



Real Chronopoli template, rendered with demo data

FIG. 1 Question 2 of 5, as rendered by the real start.html: seven options, one per district. Layla picks Compliance & AML; the gold bar shows her progress. Rendered with a Mako engine in the test harness; in the LMS the page shows as literal text until R-9 is fixed.



Real Chronopoli template, rendered with demo data

FIG. 2 Layla's result from the real results view: Compliance District as home district, the Professional layer, and Investigation and Digital Assets as neighbouring districts. Rendered as above (R-9).

How the recommendation is calculated

- **Home district:** your answer to Q2.
- **Neighbouring districts:** two complementary districts are always added. Compliance pairs with Investigation and Digital Assets.
- **Learning layer:** taken from your role (Q1) and adjusted by your experience (Q3). Beginners always start at L1; experienced professionals and senior experts never start at L1. Layla, an experienced exchange MLRO, starts at L2 Professional.
- **What you see next:** the results page shows your districts and layer only. It does not list courses; you find them in the course catalogue and in the partner academies (chapter 4).

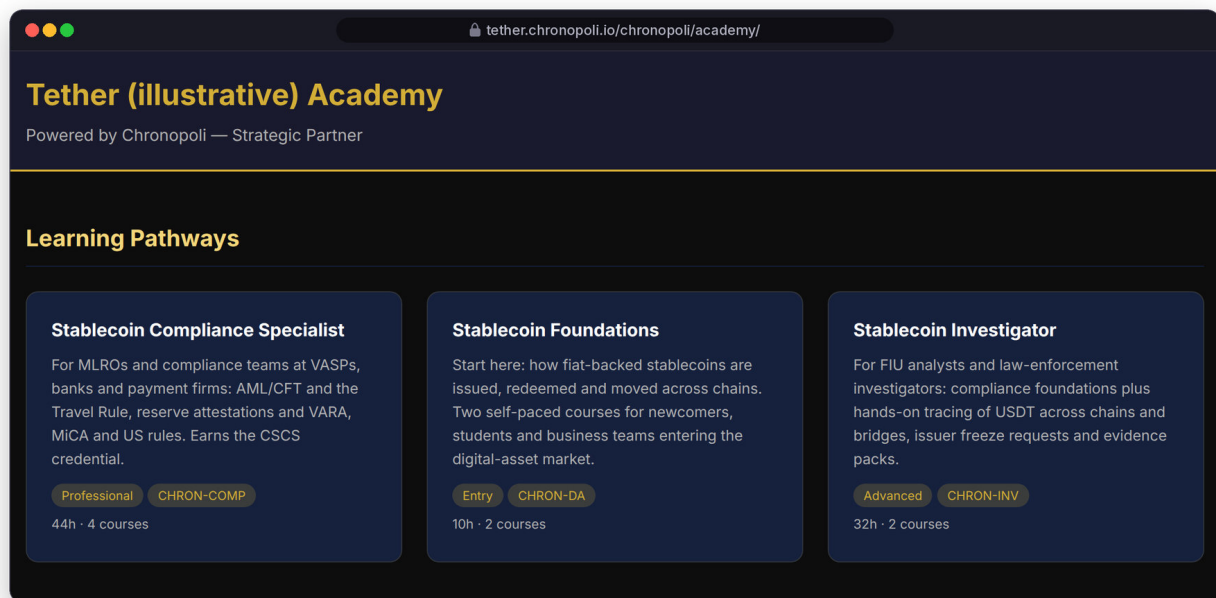
● NOTE

Your answers are stored in your OnboardingProfile. Other applications can read the result from </chronopoli/api/onboarding/status/>. The AI Tutor uses it to adjust its tone and the examples it picks, and the community sign-on uses your home district to add you to its group.

— CHAPTER 04

The Tether academy and its courses

A Company Academy is a partner's own space inside Chronopoli. It groups the partner's courses into pathways, awards co-branded badges when a pathway is finished and links those badges to open roles. The courses themselves run in Open edX.



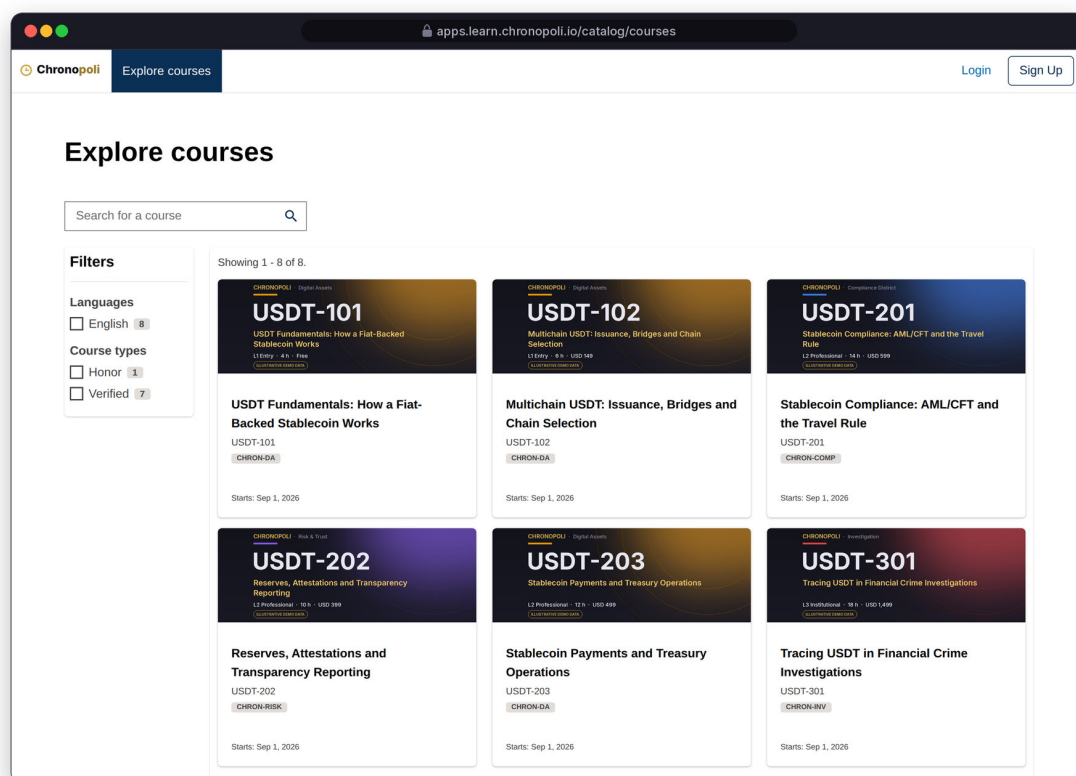
Real Chronopoli template, rendered with demo data · FIG. 3 The academy home on the partner subdomain: the real `home.html` lists the three pathways with level, district, hours and number of courses (illustrative Tether data). The page shows the partner's name; the stored brand colour and logo are not used yet, and the subdomain middleware was enabled in the test harness because the plugin does not register it today (R-23).

Learning pathways and badges (mock)

PATHWAY	LEVEL	COURSES	HOURS	BADGE AWARDED ON COMPLETION
Stablecoin Foundations	Entry	USDT-101, USDT-102	10 h	Stablecoin Foundations badge
Stablecoin Compliance Specialist	Professional	USDT-101, USDT-201, USDT-202, USDT-302	44 h	Certified Stablecoin Compliance Specialist (CSCS)
Stablecoin Investigator	Advanced	USDT-201, USDT-301	32 h	Stablecoin Investigator badge

Course catalogue (mock)

CODE	COURSE TITLE	DISTRICT	LAYER	EFFORT	PRICE
USDT-101	USDT Fundamentals: How a Fiat-Backed Stablecoin Works	DA	L1	4 h	Free
USDT-102	Multichain USDT: Issuance, Bridges and Chain Selection	DA	L1	6 h	\$149
USDT-201	Stablecoin Compliance: AML/CFT and the Travel Rule	COMP	L2	14 h	\$599
USDT-202	Reserves, Attestations and Transparency Reporting	RISK	L2	10 h	\$399
USDT-203	Stablecoin Payments and Treasury Operations	DA	L2	12 h	\$499
USDT-301	Tracing USDT in Financial Crime Investigations	INV	L3	18 h	\$1,499
USDT-302	Regulating Stablecoins: VARA, MiCA and US Frameworks	GOV	L3	16 h	\$1,299
USDT-401	Stablecoin Executive Programme (Live Cohort, Dubai)	DA	L4	5 days	\$4,500



Real Open edX screen · FIG. 4 The Open edX course catalogue with the eight USDT courses as district-coded cards (the first six are shown), full-text search and filters by language and course type: one free course in **honor** mode and seven **verified** courses (illustrative Tether data).

Every course has an Open edX course key such as `course-v1:CHRON-COMP+USDT-201+2026_Q4`. The organisation part is the district, so a Tether course appears in its district's catalogue; a PartnerTrack links it to the partner. The run name 2026_Q4 is the planned go-live label; the demo runs start on 1

September 2026, which is why the screens show that date and chapter 05 reports Q3 sales.

Inside a course: USDT-201 (mock)

Stablecoin Compliance: AML/CFT and the Travel Rule is Layla's first professional course. It has six modules. Modules 1 to 5 each pair a lesson with a short graded knowledge check, and module 6 is a graded capstone case study. The course is passed at 70 per cent, which earns a verified certificate.

MODULE	TOPIC	LEARNING OUTCOME	GRADED WORK
1	How a fiat-backed stablecoin works	Explain issuance, redemption, reserves and why the peg matters for AML risk	Knowledge check (2 questions)
2	Risk-based approach for VASPs	Build a stablecoin risk assessment across customers, chains and corridors	Knowledge check (2 questions)
3	Freeze controls and sanctions screening	Know when issuer-level freezes apply and what a VASP must still do itself	Knowledge check (2 questions)
4	The Travel Rule in practice (FATF R.16)	Apply FATF Recommendation 16 to USDT transfers between VASPs	Knowledge check (2 questions)
5	Monitoring and red flags	Spot structuring, chain-hopping and mixer exposure in stablecoin flows	Knowledge check (2 questions)
6	Capstone case study	Work through the Gulf Digital Exchange case and decide on escalation	Capstone (4 questions)

apps.learn.chronopoli.io/learning/course/course-v1:CHRON-COMP+USDT-2...

Chronopoli · Compliance District USDT-201
Stablecoin Compliance: AML/CFT and the Travel Rule

Course Progress Dates

Stablecoin Compliance: AML/CFT and the Travel Rule

Pick up where you left off [Resume course](#) [Expand all](#)

- ✓ **Module 1: How a fiat-backed stablecoin works** +
- ✓ **Module 2: Risk-based approach for VASPs** +
- ✓ **Module 3: Freeze controls and sanctions screening** +
- ✓ **Module 4: The Travel Rule in practice (FATF R.16)** +
- ✓ **Module 5: Monitoring and red flags** +
- ✓ **Module 6: Capstone case study (graded, pass >= 70%)** -
 - ✓ 6.1 Capstone: Gulf Digital Exchange (4 Questions) Capstone

Course Tools

- [Bookmarks](#)
- [Launch tour](#)

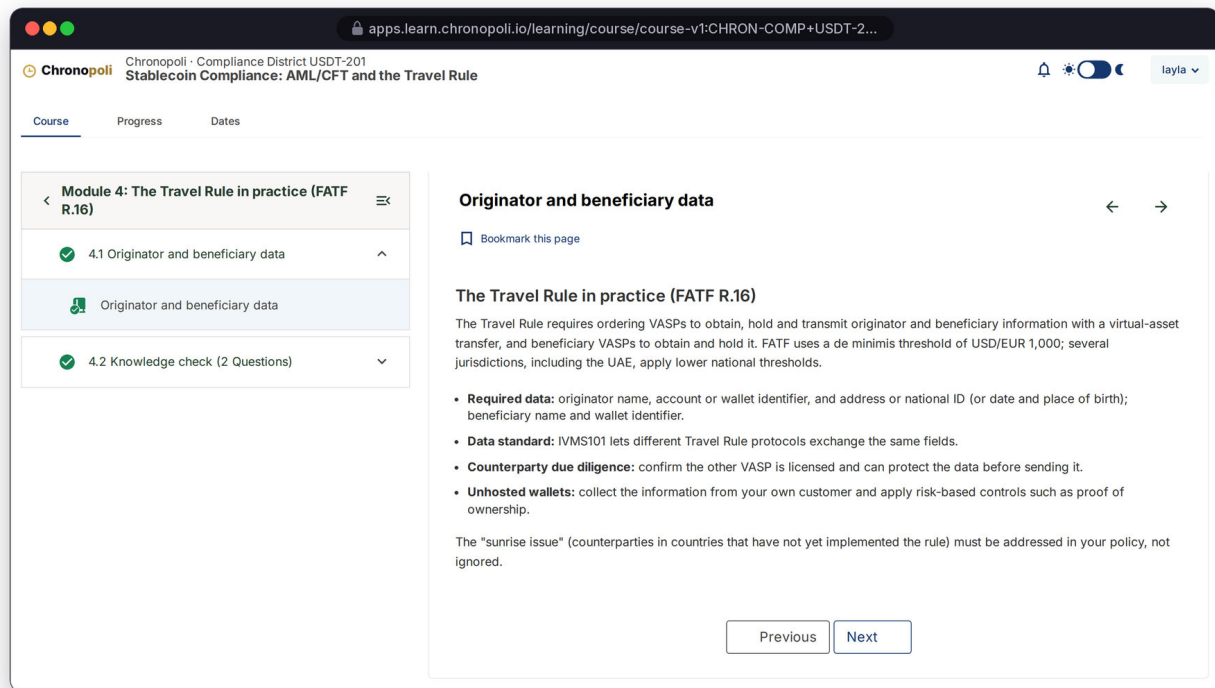
Important dates

Thu, Dec 31, 2026
Course ends
After this date, the course will be archived, which means you can review the course content but can no longer participate in graded assignments or work towards earning a certificate.
[View all course dates](#)

Course Handouts

1. FATF R.15/R.16 guidance summary (PDF)
2. IVMS101 field checklist (PDF)
3. goAML STR template (UAE) (PDF)

Real Open edX screen · **FIG. 5** The USDT-201 course home in the Open edX Learning app: all six modules completed by Layla, module 6 opened on the capstone, the course end date (31 Dec 2026) and three PDF handouts (illustrative Tether data).



Real Open edX screen · FIG. 6 Lesson 4.1, 'Originator and beneficiary data': the module navigation with completion ticks on the left and the lesson on the Travel Rule, IVMS101 and unhosted wallets on the right (illustrative Tether data).

Working through a unit

- **Navigate** with the module outline on the left or with Previous and Next. A green tick marks every unit you have completed, and Resume course on the course home takes you back to where you left off.
- **Bookmark** a unit with 'Bookmark this page'; your bookmarks are listed under Course Tools on the course home, next to the course handouts (for USDT-201 the FATF summary, the IVMS101 field checklist and a goAML report template).
- **Knowledge checks** are graded. In USDT-201 each question allows three attempts, and the last answer you submit counts (Open edX calls this 'Last Score').
- **Progress and Dates:** the Progress tab shows your completion and weighted grade (chapter 7); the Dates tab lists deadlines such as the course end on 31 December 2026.

CHAPTER 05

Enrolment, pricing and payment

Free courses open with one click. Paid courses use Stripe Checkout. Organisations can buy seat-based team licences, and partner revenue is split automatically.

Stablecoin Compliance: AML/CFT and the Travel Rule
Chronopoli · Compliance District

AML/CFT, sanctions screening and FATF Travel Rule controls for VASPs handling stablecoins. 6 modules, graded capstone.

[Enroll now](#)

About this course

Stablecoin Compliance: AML/CFT and the Travel Rule is part of the **Tether Stablecoin Infrastructure Academy** on Chronopoli, hosted in the Compliance District (CHRON-COMP), learning layer L2 (Professional).

Effort: 14 h. Price: USD 599.

Illustrative demo course. Tether/USDT course names, people, organisations and figures are fictitious mock data; Tether is on the partner roadmap only and no partnership is implied.

Modules

1. How a fiat-backed stablecoin works
2. Risk-based approach for VASPs
3. Freeze controls and sanctions screening
4. The Travel Rule in practice (FATF R.16)
5. Monitoring and red flags
6. Capstone case study (graded, pass $\geq 70\%$)

USDT-201
Stablecoin Compliance: AML/CFT and the Travel Rule
L2 Professional · 14 h · USD 599
(ILLUSTRATIVE DEMO DATA)

[X](#) [f](#) [✉](#)

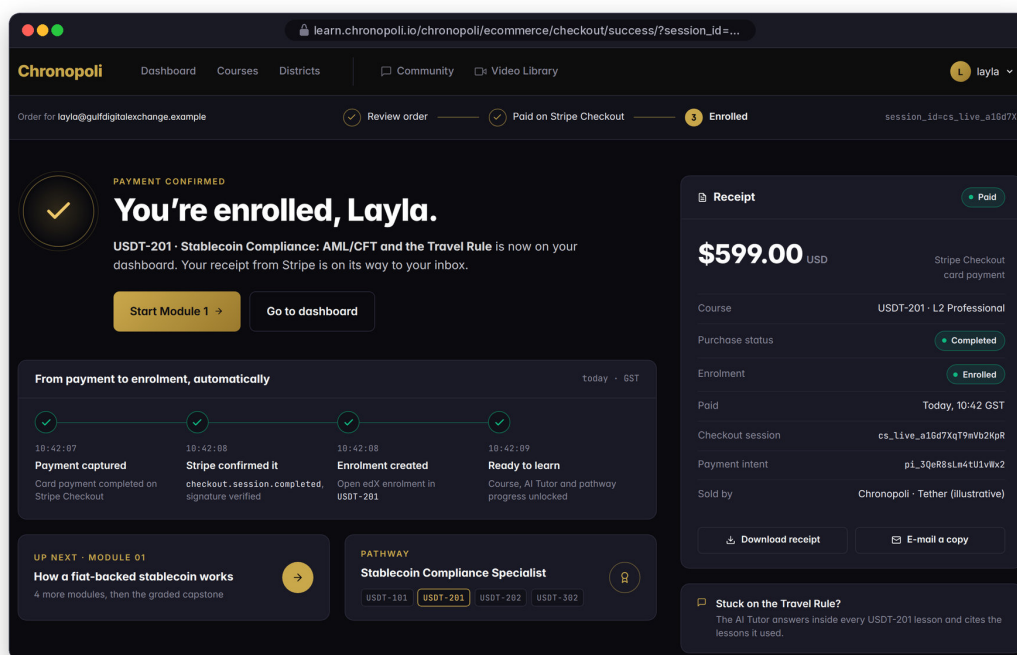
Course number	USDT-201
Classes start	Sep 1, 2026
Classes end	Dec 31, 2026
Estimated effort	14 h
Price	\$599

Real Open edX screen · FIG. 7 The USDT-201 course page in the Open edX catalogue: summary, the six-module syllabus, dates, 14 hours of effort and the price of the verified track, USD 599 (illustrative Tether data).

- 1 Choose 'Enroll now' on the course page**
 For free courses such as USDT-101, you are enrolled straight away. Paid courses open the Chronopoli checkout for the verified track (target behaviour; until R-12 is closed the standard button enrolls without payment, see the IMPORTANT note after these steps).
- 2 Pay with Stripe Checkout**
 Stripe takes the payment by card and, on supported devices, Apple Pay or Google Pay. Chronopoli never stores card details.
- 3 Automatic enrolment**
 A signed Stripe webhook (checkout.session.completed) marks the purchase as paid and enrolls you in Open edX (target behaviour; refund handling and webhook hardening are still open, R-14).
- 4 Receipt and revenue split**
 Stripe e-mails your receipt. For partner courses, 70 per cent of the payment is recorded as a payout to the partner's Stripe Connect account (the payout status is not yet updated from Stripe, R-13).

● IMPORTANT

Until finding R-12 is closed, the standard Open edX enrol button and API do not check for payment. Even with the verified modes shown in Fig. 19, learners can enrol in audit for free (Fig. 21). Removing the audit mode is not enough on its own (Fig. 19 already has none), because Open edX still falls back to audit. An openedx-filters enrolment step that rejects unpaid enrolments in verified-only courses must be in place before paid courses go live.



Design prototype – template not yet built (R-10) · FIG. 8 After payment: the planned success page with the receipt, the payment-to-enrolment timeline and the next module (illustrative Tether data). The template ecommerce/success.html is missing today.

Revenue split on partner courses

Individual purchases of Tether courses in Q3 2026 (mock). Seats paid for through team licences are not included.

COURSE	PAID ENROLMENTS	LIST PRICE	GROSS	PARTNER 70%	PLATFORM 30%
USDT-102	380	\$149	\$56,620	\$39,634	\$16,986
USDT-201	520	\$599	\$311,480	\$218,036	\$93,444
USDT-202	260	\$399	\$103,740	\$72,618	\$31,122
USDT-203	170	\$499	\$84,830	\$59,381	\$25,449
USDT-301	140	\$1,499	\$209,860	\$146,902	\$62,958
USDT-302	90	\$1,299	\$116,910	\$81,837	\$35,073
USDT-401	50	\$4,500	\$225,000	\$157,500	\$67,500
Total	1,610		\$1,108,440	\$775,908	\$332,532

● NOTE

Cancelled checkouts leave no enrolment and no partner payout. Every Stripe event is stored once, keyed by its event ID, so a webhook that Stripe sends twice cannot enrol a learner twice.

Team licences

Companies buy seats rather than single courses. The default list price is **\$125 per seat per month**. An administrator assigns seats to named learners and can reassign a seat when someone leaves. Layla's employer, Gulf Digital Exchange, holds 25 seats and has assigned 18 of them. Layla bought USDT-201 herself before GDX took out the licence on 1 September; the seats are for her colleagues, and the licence is managed by GDX's learning and development team.

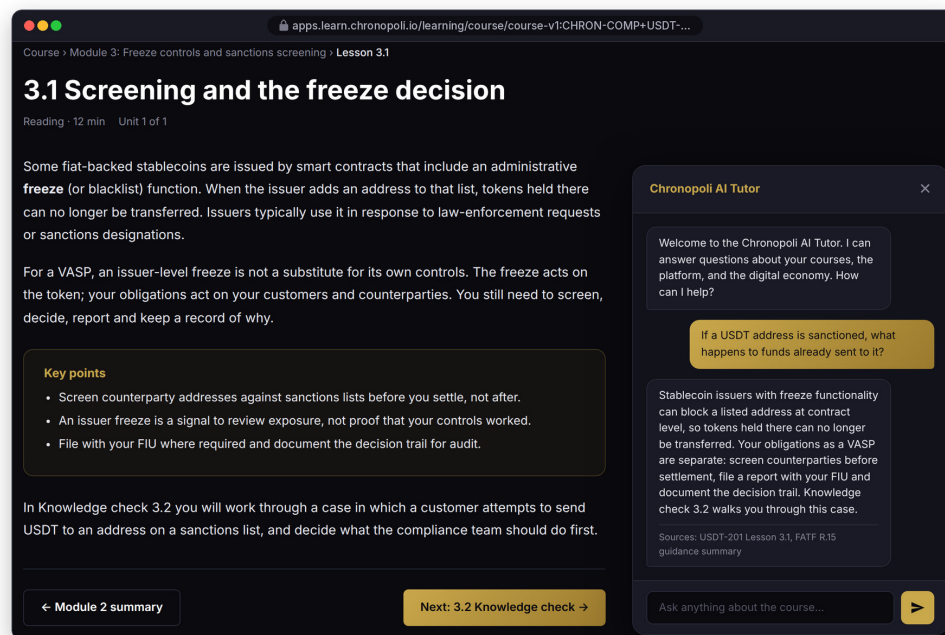
The screenshot displays the 'Team licences' interface for Gulf Digital Exchange (GDX). At the top, it shows the account name and a link to invoices. The main section features a large '18 / 25 seats assigned' indicator with a visual bar showing 18 assigned seats (yellow) and 7 available seats (grey). A progress bar indicates '72% of licensed seats in use'. To the right, a 'MONTHLY PRICE' section shows 'Price per seat' at \$125/month and 'Licensed seats' at 25, resulting in a 'Total per month' of \$3,125. Below this, a table lists key details: Current Period (1 Sep - 30 Sep 2026), Next Invoice (1 Oct 2026 - \$3,125.00), Billing (Stripe Billing - monthly), Stripe Customer ID, and Subscription ID. The bottom section lists three course categories: ENTRY (Stablecoin Foundations), PROFESSIONAL (Stablecoin Compliance Specialist), and ADVANCED (Stablecoin Investigator), each with specific course codes and durations. A 'How the licence works' sidebar explains that one seat equals one named learner, seats can be reassigned, and adding seats updates the Stripe subscription quantity.

Design prototype – template not yet built (R-10) · FIG. 9 The planned team-licence page, signed in as GDX's learning and development billing contact: 18 of 25 seats assigned, $\$125 \times 25 = \$3,125$ per month, the billing period and the pathways the seats unlock (illustrative Tether data; Gulf Digital Exchange is fictitious). The template `ecommerce/subscriptions.html` is missing today.

CHAPTER 06

Learning with the AI Tutor

The AI Tutor is a chat assistant on the course pages. It answers from the Chronopoli knowledge base and cites its sources, so you can check every answer. It gives no legal, investment or compliance advice.



Design prototype – real widget code on a composed lesson page · FIG. 10 Lesson 3.1, while Layla is still working towards her CSCS badge: she asks about a sanctioned USDT address; the answer cites 'USDT-201 Lesson 3.1' and a 'FATF R.15 guidance summary' (illustrative Tether data). Not yet verified inside the Learning app.

What the tutor knows about you

- Your home district and layer from onboarding. This sets the depth and the examples it uses.
- The course and unit you are viewing, so that short questions make sense.
- Your earlier tutor sessions, which the tutor keeps as your conversation history.

Getting good answers

INSTEAD OF ...	TRY ...	WHY
"Explain stablecoins"	"Explain how USDT redemption works for an exchange that is not a direct Tether customer"	A specific role and scenario gives a specific answer
"Is this allowed?"	"What does the VARA rulebook require when a USDT transfer arrives with incomplete Travel Rule data?"	Naming the rulebook points the tutor to the right sources
"Quiz me"	"Give me five scenario questions on module 3 at L2 level, then mark my answers"	Structured practice that fits your layer

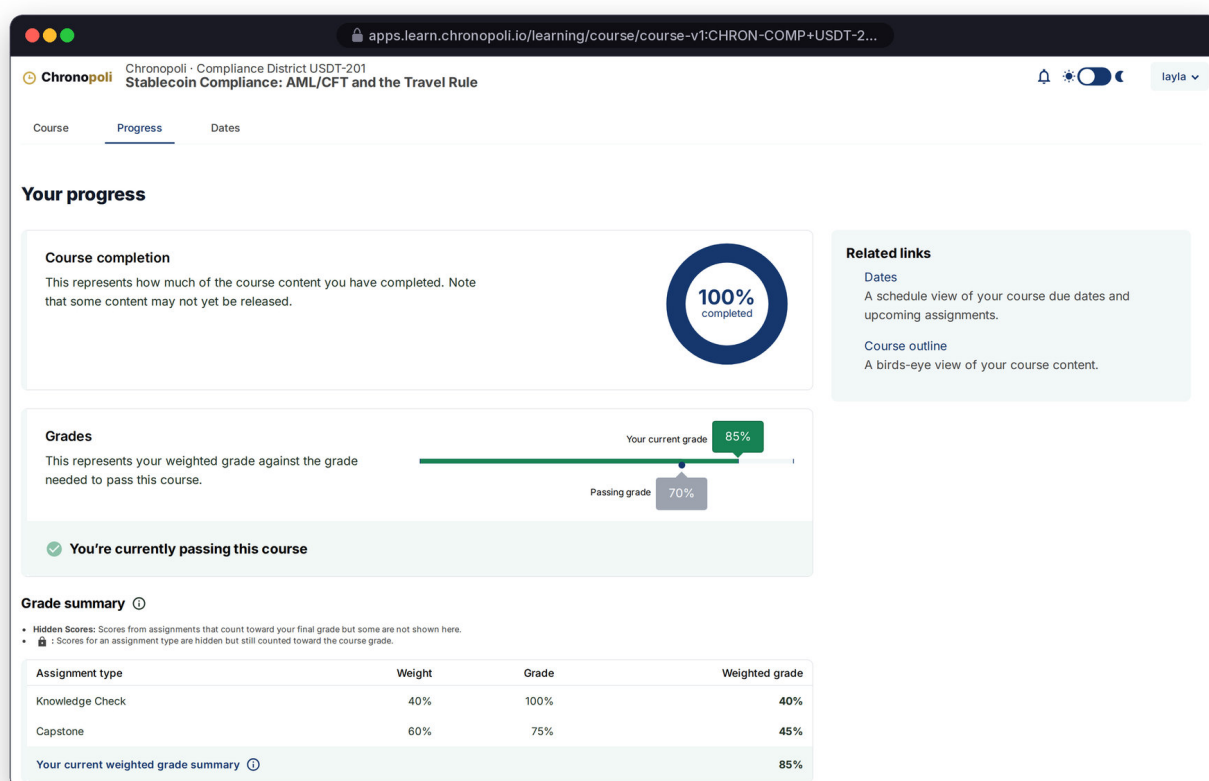
CHAPTER 07

Grades, badges and careers

Passing a paid course earns a verified certificate; free courses count towards pathway badges. Finishing a pathway earns a verifiable Open Badge, and partners can link open roles to badges, so your credentials reach the teams that are hiring.

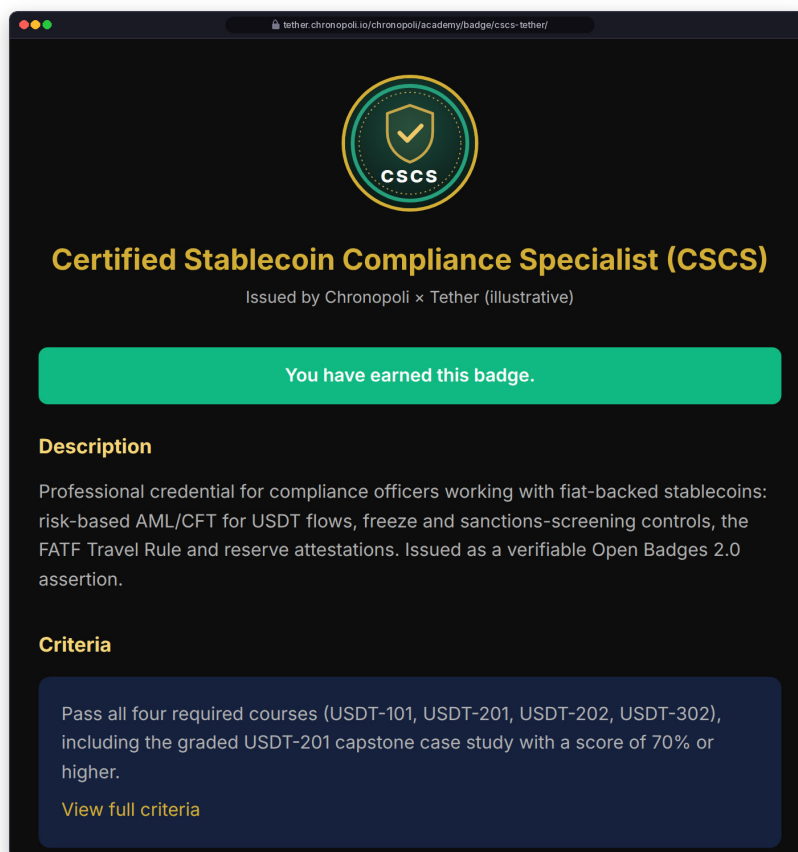
Your grade

Open edX weights the graded work of each course. In USDT-201 the knowledge checks count for 40 per cent and the capstone for 60 per cent. Layla answered every knowledge-check question correctly and three of the four capstone questions, so she passes with 85 per cent against a pass mark of 70 per cent.



Real Open edX screen · FIG. 11 Layla's progress page for USDT-201: 100 per cent complete, a weighted grade of 85 per cent against the 70 per cent pass mark, and the grade summary ($40\% \times 100\% + 60\% \times 75\%$) (illustrative Tether data).

Your badge



Real Chronopoli template, rendered with demo data · FIG. 12 The CSCS badge page from the real badge.html, as Layla sees it (illustrative Tether data).

On 18 September 2026 Layla completes the last of the four required courses of the **Stablecoin Compliance Specialist** pathway (USDT-101, USDT-201, USDT-202 and USDT-302) and receives the **Certified Stablecoin Compliance Specialist (CSCS)** badge.

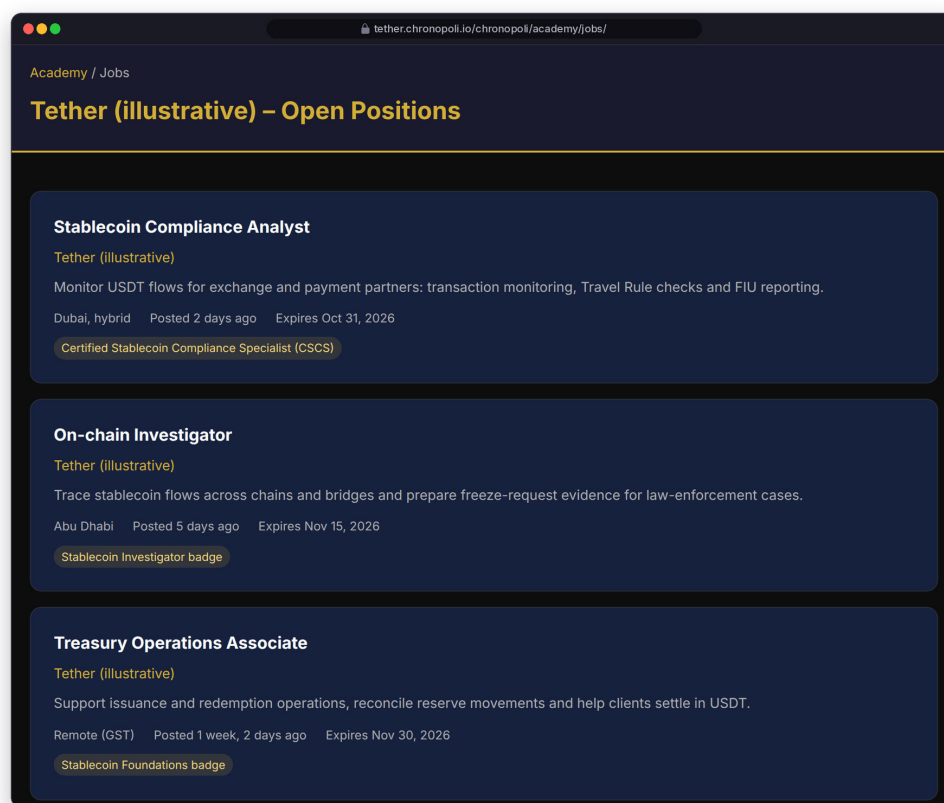
The badge page names the issuer, confirms that she has earned the badge and lists the criteria. The page itself does not show the issue date, which is kept with the award record.

The badge follows the Open Badges 2.0 standard, which LinkedIn, badge wallets and HR systems can read. The assertion below shows the target data format; salted recipient hashes and consistent assertion IDs are part of finding R-37.

OPEN BADGES 2.0 ASSERTION (TARGET FORMAT, MOCK)

```
{
  "@context": "https://w3id.org/openbadges/v2",
  "type": "Assertion",
  "id": "https://tether.chronopoli.io/chronopoli/academy/api/assertions/5b0e...c41",
  "recipient": {
    "type": "email", "hashed": true, "salt": "d41f...9a", "identity": "sha256$9f2c..."
  },
  "badge": "https://tether.chronopoli.io/chronopoli/academy/badge/cscs-tether/",
  "issuedOn": "2026-09-18T10:42:00Z",
  "verification": { "type": "hosted" }
}
```

Open roles linked to badges (mock)



Real Chronopoli template, rendered with demo data · FIG. 13 The academy's open positions from the real jobs.html, each with the badge it asks for (illustrative Tether data). A fourth role, Partnerships Manager, MENA, follows further down the page.

ROLE	LOCATION	REQUIRED BADGE	APPLICANTS
Stablecoin Compliance Analyst	Dubai, hybrid	CSCS	21
On-chain Investigator	Abu Dhabi	Stablecoin Investigator	9
Treasury Operations Associate	Remote (GST)	Stablecoin Foundations	5
Partnerships Manager, MENA	Dubai	Any Tether (illustrative) academy badge	2

Applicant counts come from the talent pipeline; their total (37) appears on the partner dashboard in chapter 11. The jobs page itself does not show them.

1

Opt in

In your profile, allow partners to see your badges. You can withdraw at any time. (Today the talent pipeline does not ask for this consent; closing finding R-37 in Part IV, planned for sprint 2, adds it.)

2

Automatic matching

When a badge is issued, the talent pipeline checks it against active roles that require that badge.

3

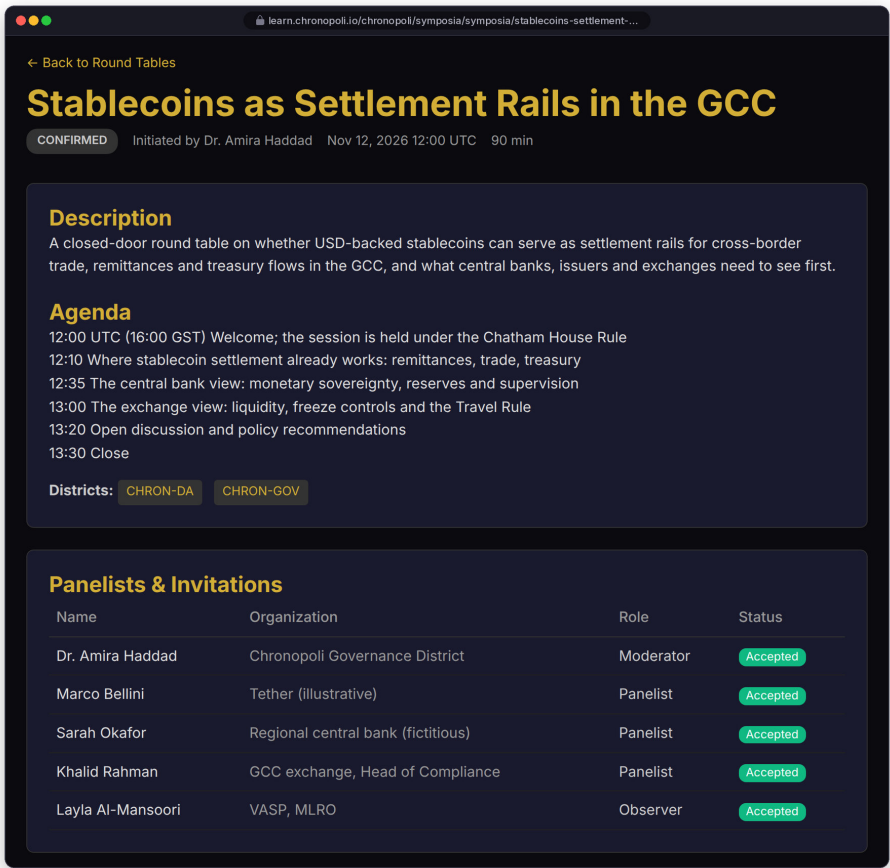
Apply

A snapshot of your credentials is submitted and the partner's HR contact is notified by e-mail.

CHAPTER 08

Symposia round tables

Round tables are closed expert sessions, streamed live to registered participants and held under the Chatham House Rule. After each session an AI pipeline turns the recording into a transcript, an analysis and a set of publications, and a person approves every output before it goes out.



Real Chronopoli template, rendered with demo data · FIG. 14 The round table 'Stablecoins as Settlement Rails in the GCC' before the event: status Confirmed, 12 Nov 2026 with the agenda in UTC (12:00 UTC is 16:00 GST), districts CHRON-DA and CHRON-GOV, and the accepted panel with Layla as observer (illustrative Tether data). The doubled /symposia/symposia/ in the address is finding R-21.

Lifecycle of a round table

STATUS	WHAT HAPPENS	WHO ACTS
Draft	Title, districts, agenda and date are prepared	Initiator
Open for invitations	Invitations go out with a secure token and an expiry date	Moderator
Confirmed	Enough panellists have accepted; an Amazon IVS channel is created	Platform
Live / Recording	The session is streamed and recorded to S3	Panel
Processing	Step Functions runs transcription, analysis and the output generators	Pipeline
Completed	Every output is waiting for review in the approval queue	Moderator
Published	Approved outputs are released to learners, partners and social channels	Moderator

What the pipeline produces

Transcript

Full text from Amazon Transcribe, with speakers separated.

Analysis

Key themes, points of agreement and dissent, and open questions.

Opinion paper

A PDF opinion paper in Chronopoli design, with no names attached (Chatham House).

Partner report

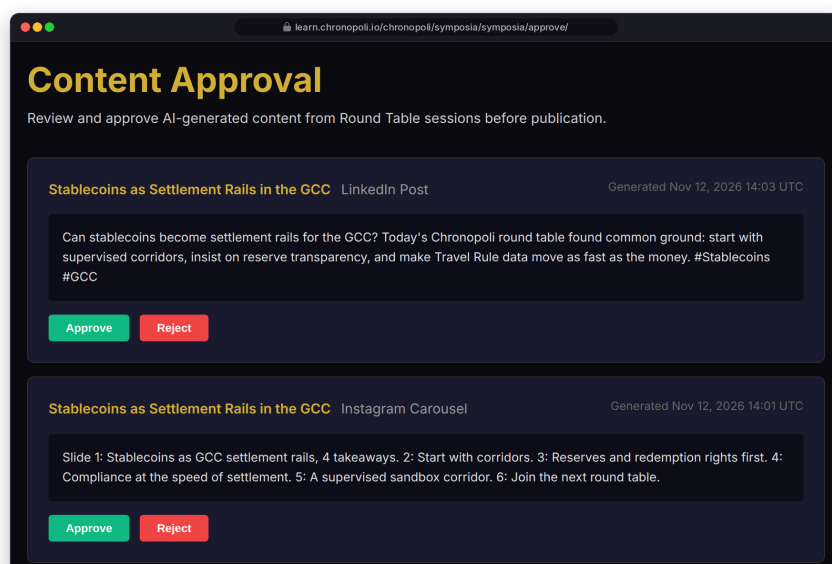
A briefing for the sponsoring partner, here Tether (illustrative).

LinkedIn post

A ready-to-post summary for the district's channel.

Instagram carousel

Slide texts and a visual brief for a carousel.



● KEY POINT

Nothing is published automatically. Dr. Haddad approves or rejects each output in the approval queue before it goes out.

Until R-21 is closed, the queue's address repeats the symposia segment (see the URL map in chapter 13).

Real Chronopoli template, rendered with demo data · FIG. 15 The staff approval queue from the real approve.html, filled with simulated outputs for the 12 Nov session, so the timestamps on screen show that future date (illustrative Tether data).

— CHAPTER 09

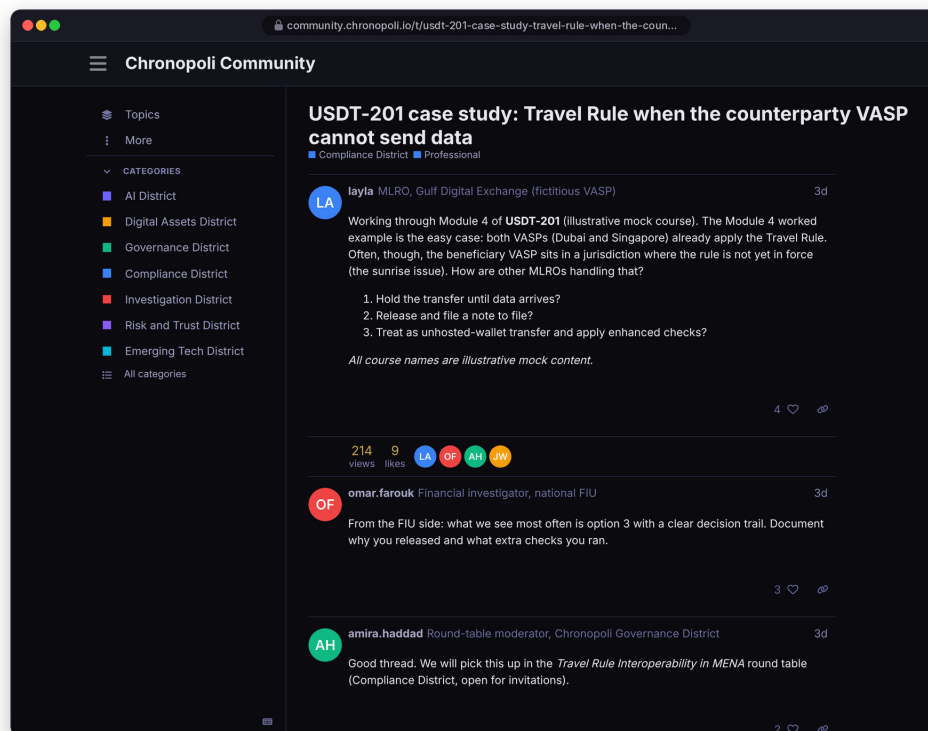
Community, video and slides

Three companion services extend the LMS. You reach them from the top navigation; the community signs you in with your Chronopoli account.

AREA	ADDRESS	WHAT YOU DO THERE	BUILT ON
Community	community.chronopoli.io	Discuss in district categories and course threads	Discourse + Chronopoli SSO
Video	video.chronopoli.io	Watch lecture recordings and round-table replays	Opencast
Slides	slides.chronopoli.io	Instructors generate district-branded slide decks with AI	Presenton

District groups in the community

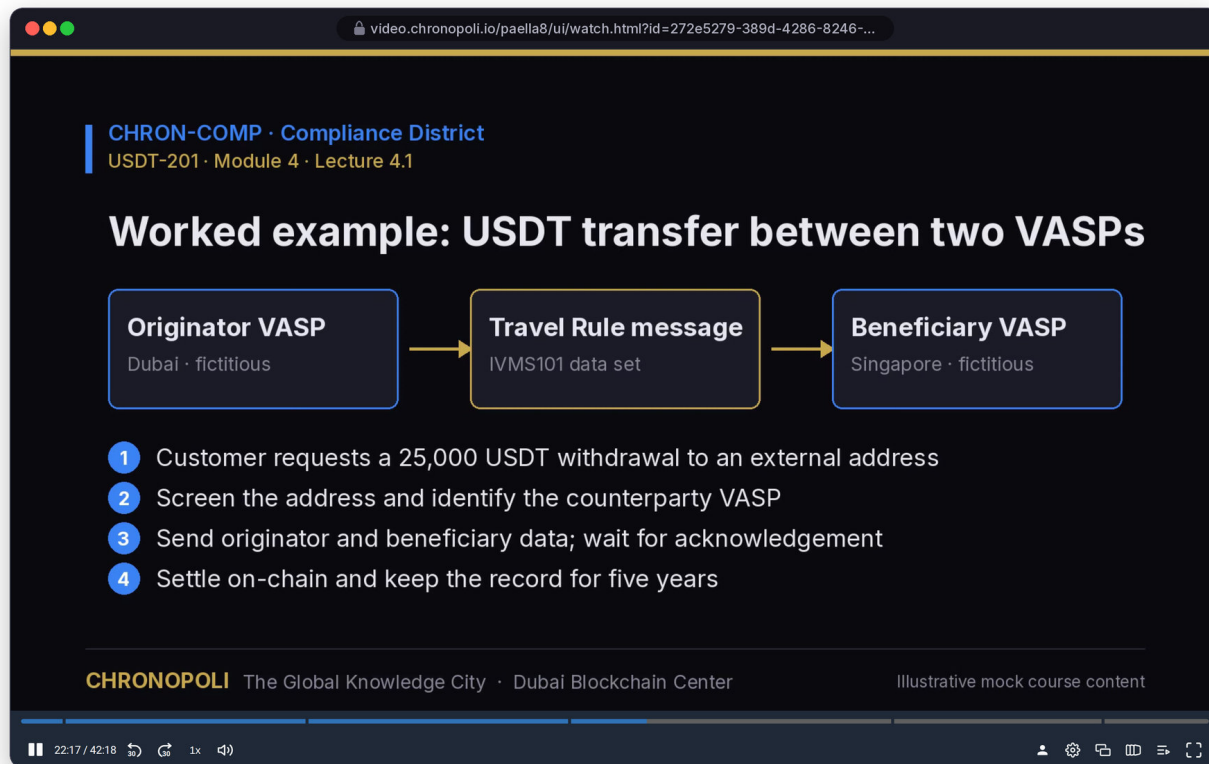
When you first sign in to the community, the SSO handshake adds you to the group for your home district. Layla joins **compliance-district**. Each district has its own category with Entry Level, Professional, Institutional and Partner Tracks subcategories, and course threads go in the subcategory for the course's layer (USDT-201 threads sit under Compliance District > Professional).



Real extension (Discourse) · FIG. 16 A course-linked thread on Discourse 3.5: Layla, while still working through module 4, asks how other MLROs handle the Travel Rule sunrise issue, and an FIU investigator and the round-table moderator reply. All personas are fictitious (illustrative Tether data).

Video lectures

Lectures and round-table replays are processed by Opencast. It publishes each recording to the Paella player, detects the slides and builds a clickable slide index. Every recording carries its district, course series and rights metadata.



Real extension (Opencast) · **FIG. 17** Lecture 4.1 of USDT-201 in the Paella 8 player of Opencast 20.4, at 22:17 of 42:18, with the slide-segment markers on the timeline (illustrative Tether data).

Slides for instructors

Presenton uses per-district prompt templates. The template sets colours, tone and slide structure. An instructor for the illustrative Tether academy can ask for "a 12-slide L2 lesson on Travel Rule data fields for USDT transfers" and receive a deck in Compliance District design that can be edited further. For now the Slides builder sits behind a separate staff login; single sign-on with the Chronopoli account is planned for Phase 2.

03

PART THREE

Partner & admin guide

How a partner academy is run, shown with the illustrative Tether academy: set-up, dashboards, Digital Twin masterclasses and a quick reference for operators.

IN THIS PART

10	Running a partner academy	27
11	Partner dashboard and intelligence	29
12	Digital Twin masterclasses	31
13	Administrator quick reference	32

— CHAPTER 10

Running a partner academy

This chapter is for partner administrators like Nadia. It explains how to set up an academy, publish courses and pathways and get paid. Most set-up steps happen in the Django admin at /admin/, and course content is built in Open edX Studio.

- 1 Create the Partner record**
Enter the name, tier, org code, districts and contact. Chronopoli staff do this during partner onboarding.
- 2 Enable the academy**
Set the subdomain to tether and tick academy_enabled. The subdomain middleware then serves the academy on tether.chronopoli.io once it is registered (R-23).
- 3 Connect payouts**
Add the Stripe Connect account ID. Payouts are created when a purchase is completed.
- 4 Build the course in Studio**
Create the course under the district organisation and add modules, lessons and graded checks.
- 5 Set the price**
Add a verified course mode with the USD price and upgrade deadline, then link the course to the partner with a PartnerTrack.
- 6 Build pathways and badges**
Create a LearningPathway with ordered PathwaySteps (course keys) and attach a Badge with its criteria, image and issuer.
- 7 Post roles**
Create PartnerJobPostings with the required badges and an expiry date. Applications arrive in the admin and by e-mail.

● NOTE

Academy pages show the partner's name ('Tether (illustrative) Academy') and otherwise use the Chronopoli gold and navy palette. The brand colour and logo fields on the Partner record are stored but not yet applied by the academy templates. The rest of the platform always keeps the Chronopoli design. The planned pages (Fig. 9 and Fig. 20) also show the academy's own title, here Tether Stablecoin Infrastructure Academy (illustrative).

The screenshot shows the Chronopoli LMS interface for a course titled "Stablecoin Compliance: AML/CFT and the Travel Rule". The course is active from Sep 01, 2026 to Dec 31, 2026, and is 4/7 completed. The course outline lists four modules:

- Module 1: How a fiat-backed stablecoin works
- Module 2: Risk-based approach for VASPs
- Module 3: Freeze controls and sanctions screening
- Module 4: The Travel Rule in practice (FATF R.16)

Each module has a "Section highlights" link. The right sidebar shows course details, including a "Course Content Summary" with 49 total items, 6 videos, 6 quizzes, 14 assignments, and 11 tests. It also shows "Taxonomy Alignments".

Real Open edX screen · FIG. 18 USDT-201 in the Open edX Studio course outline: the course is active from 1 Sep to 31 Dec 2026, and modules 1 to 4 of 6 are visible with the content summary on the right (illustrative Tether data).

The screenshot shows the LMS Administration interface for "course_modes/coursemode/". It displays a table of course modes with columns: ID, COURSE, MODE, PRICE, UPGRADE DEADLINE, SKU, ANDROID SKU, IOS SKU, and BULK SKU. The table lists 8 course modes, including verified and honor modes.

ID	COURSE	MODE	PRICE	UPGRADE DEADLINE	SKU	ANDROID SKU	IOS SKU	BULK SKU
16	course-v1:CHRON-DA+USDT-401+2026_Q4	verified	4500	November 09, 2026, 12:00 PM	CHRON-USDT-401	-	-	-
14	course-v1:CHRON-GOV+USDT-302+2026_Q4	verified	1299	December 21, 2026, 12:00 PM	CHRON-USDT-302	-	-	-
12	course-v1:CHRON-INV+USDT-301+2026_Q4	verified	1499	December 21, 2026, 12:00 PM	CHRON-USDT-301	-	-	-
10	course-v1:CHRON-DA+USDT-203+2026_Q4	verified	499	December 21, 2026, 12:00 PM	CHRON-USDT-203	-	-	-
8	course-v1:CHRON-RISK+USDT-202+2026_Q4	verified	399	December 21, 2026, 12:00 PM	CHRON-USDT-202	-	-	-
6	course-v1:CHRON-COMP+USDT-201+2026_Q4	verified	599	December 21, 2026, 12:00 PM	CHRON-USDT-201	-	-	-
4	course-v1:CHRON-DA+USDT-102+2026_Q4	verified	149	December 21, 2026, 12:00 PM	CHRON-USDT-102	-	-	-
2	course-v1:CHRON-DA+USDT-101+2026_Q4	honor	0	-	-	-	-	-

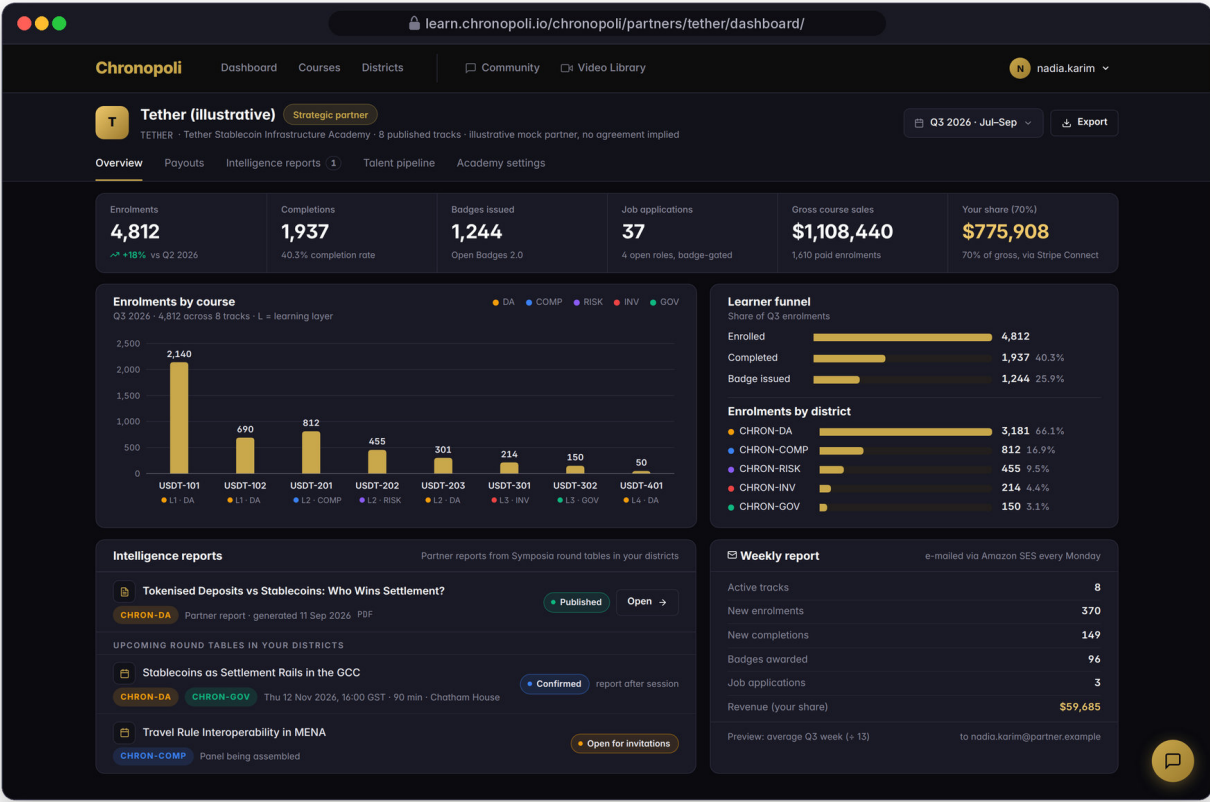
8 course modes

Real Open edX screen · FIG. 19 Prices live in the LMS back office as course modes: one **verified** mode per paid course with its USD price, upgrade deadline and SKU, and the free USDT-101 in **honor** mode. Paid courses have no free mode (illustrative Tether data).

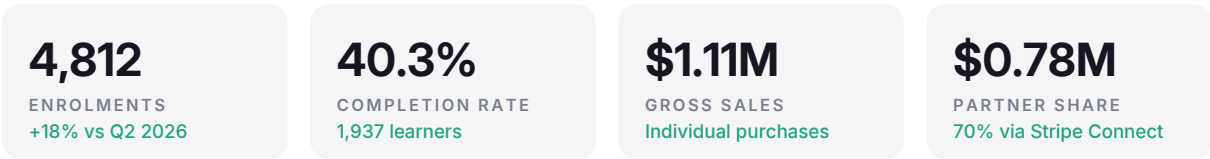
CHAPTER 11

Partner dashboard and intelligence

Partners see how their academy is doing: enrolments, completions, badges, revenue and reports built from the round tables in their districts.

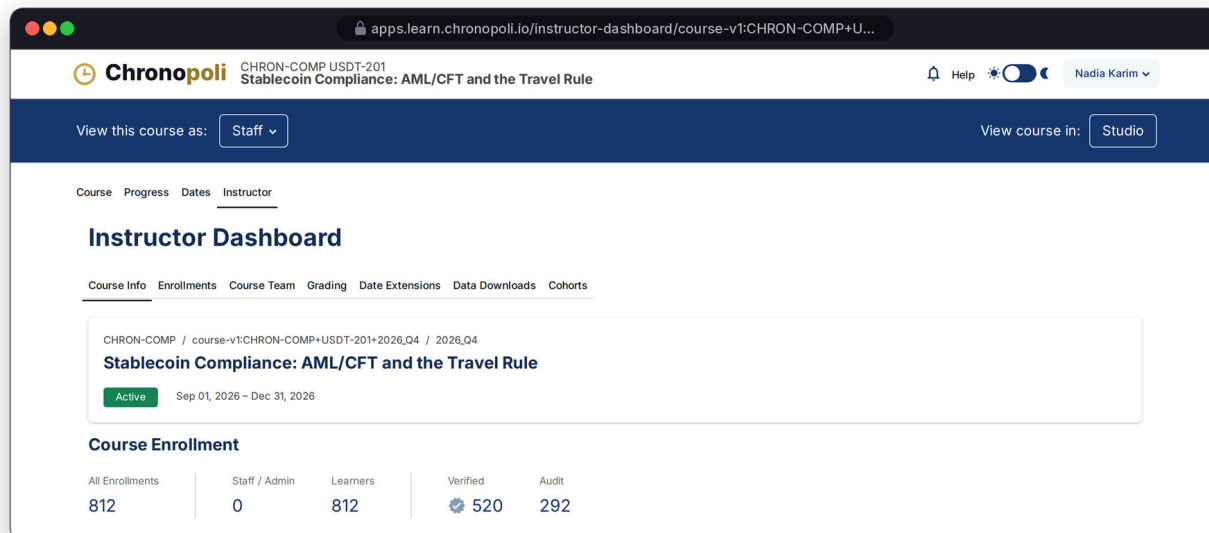


Design prototype – template not yet built (R-10) · FIG. 20 The planned Tether partner dashboard for Q3 2026, signed in as Nadia: KPI tiles, enrolments by course and district, the learner funnel, intelligence reports and the weekly report preview (illustrative Tether data). The template partners/dashboard.html does not exist yet.



What works today

Until the dashboard template exists, partner staff with a course-staff role can use the standard Open edX Instructor Dashboard for each course: enrolments by track, the course team, grading and data downloads.



Real Open edX screen · **FIG. 21** The Instructor Dashboard of USDT-201 as Nadia sees it: 812 enrolments, of which 520 are verified (paid) and 292 audit. USDT-201 has no free mode (Fig. 19), so audit enrolments here come through the standard button without payment (the gap of finding R-12) or from team-licence seats, which are not counted as individual purchases (illustrative Tether data).

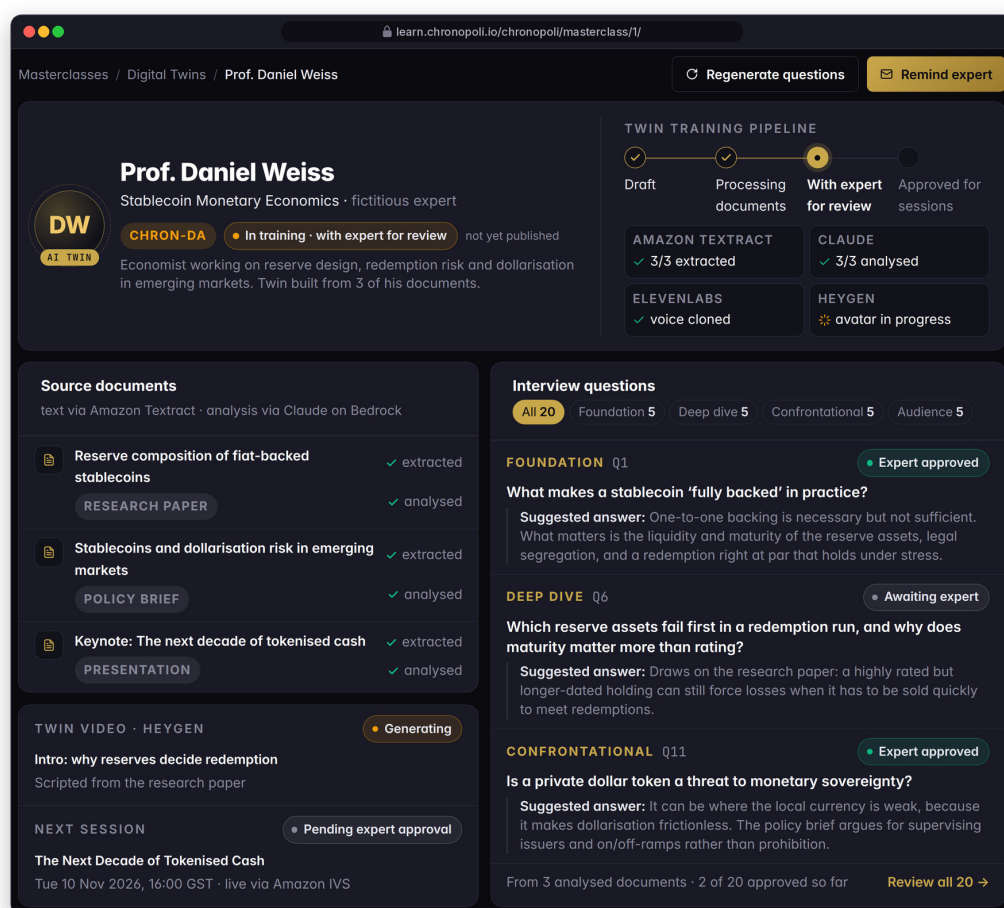
The weekly report

Every Monday morning a scheduled Celery task builds the partner report, and Amazon SES e-mails it to the partner contact: new enrolments, completions, badges, job applications, active tracks and revenue for the past seven days. The same data is available as JSON from </chronopoli/partners/api/tether/analytics/>. The task was added on the review branch; Tutor still needs a Celery beat process to run it (R-7, planned for sprint 1).

CHAPTER 12

Digital Twin masterclasses

A Digital Twin is an AI version of a real expert. It teaches only with the expert's approval, which can be withdrawn at any time.



Design prototype – template not yet built (R-10) · FIG. 22 The planned staff view of a twin under review by Prof. Daniel Weiss (fictitious expert): sources read by Textract and analysed by Claude, interview questions with suggested answers. No masterclass templates yet; unpublished twins are still visible to learners (R-30).

STEP	WHAT HAPPENS	SERVICE USED
Upload sources	Papers, policy briefs, book chapters and slides are uploaded	S3 + Amazon Textract
Analyse	Arguments, positions and vocabulary are extracted into a profile	Claude on Bedrock
Interview prep	Foundation, deep-dive, confrontational and audience questions	Claude on Bedrock
Voice and avatar	Created only with the expert's written consent	ElevenLabs, HeyGen
Live session	Streamed like a round table and can be linked to a symposium	Amazon IVS

Target workflow: the upload view is a stub today, multi-page PDFs fail in Textract (R-30) and live sessions depend on R-24.

— CHAPTER 13

Administrator quick reference

Where to find things and how to run routine operations.

URL map

PATH	PURPOSE	AUDIENCE
/chronopoli/onboarding/	Five-question onboarding and results	Learners
/chronopoli/partners/	Partner directory and partner pages	Public
/chronopoli/partners/<slug>/dashboard/	Partner KPI dashboard (no template, R-10)	Partner staff
/chronopoli/ecommerce/checkout/<course_key>/	Stripe checkout for one course	Learners
/chronopoli/ecommerce/webhook/stripe/	Stripe webhook endpoint (signed)	Stripe
/chronopoli/academy/	Academy home, pathways, badges, jobs (also on the partner subdomain)	Learners
/chronopoli/symposia/...	Round-table list, detail, schedule and approval (today under /chronopoli/symposia/symposia/, R-21)	Members, moderators
/chronopoli/ai-tutor/chat/	AI Tutor chat API used by the widget	Learners
/chronopoli/masterclass/	Digital Twin list, detail and uploads	Experts, staff
/auth/discourse/sso/	Discourse SSO handshake	Community
/admin/course_modes/coursemode/	Course prices and upgrade deadlines	Staff

Routine commands

OPERATIONS

```
# start the platform, follow the LMS logs, stop it (Phase 1, Docker on EC2)
tutor local start -d
tutor local logs -f lms
tutor local stop
# Django shell inside the LMS container
tutor local exec lms ./manage.py lms shell
# deploy, health check and backup
./scripts/deploy.sh production
./scripts/healthcheck.sh          # 8-category smoke test
./scripts/backup.sh              # databases + media to S3
```



PART FOUR

Technical review

Architecture, how the review was done, the third-party stack and its versions, 37 verified findings, the updates applied and the road to go-live.

IN THIS PART

14	Architecture overview	34
15	Third-party stack and versions	36
16	Review findings	38
17	Updates applied and road to go-live	42
18	Glossary	44

CHAPTER 14

Architecture overview

Chronopoli extends Open edX without forking it. A Tutor plugin injects settings, Django apps add the product features, and managed AWS services handle AI, media and the round-table pipeline.

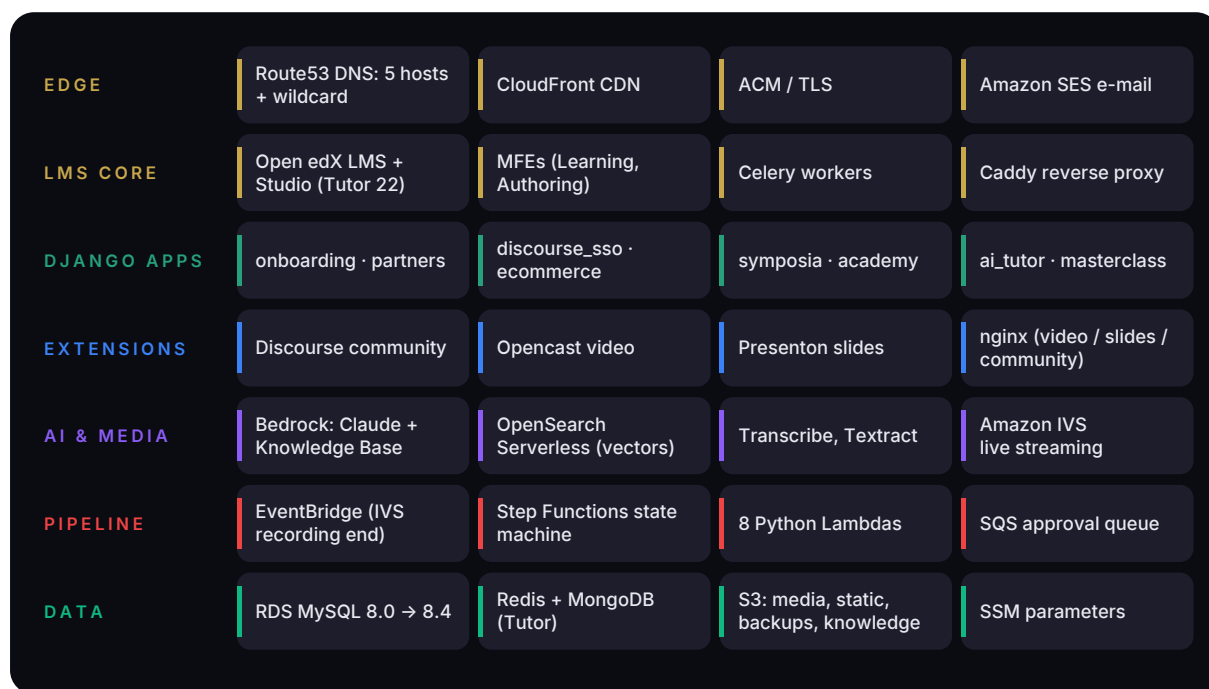


Diagram · FIG. 23 Platform layers (target design). Phase 1 runs on one EC2 host; Phase 2 moves the LMS core to EKS with autoscaling.

How the pieces connect

This is the **target design**. Where the current code does not yet work this way, the finding is named in brackets.

- **Tutor plugin** (tutorchronopoli): config defaults and generated secrets, settings patches for LMS and CMS, and an init task that creates the seven district organisations (registered on the review branch, R-8). The apps still have to be installed and routed as edx-platform plugins (R-1, R-2).
- **Django apps** hook into Open edX through signals in AppConfig.ready(): REGISTER_USER creates the onboarding profile, CourseEnrollment post_save updates purchases, and certificate events award badges (target; the certificate signal import is broken today, R-22).
- **Celery** runs symposia and masterclass work plus the weekly partner report (task added on the review branch; Tutor still needs a beat process, R-7, planned for sprint 1).
- **Round-table pipeline**: IVS recording end → EventBridge → Step Functions (Transcribe, Claude analysis, four parallel generators, notify) → SQS → Django approval queue → task-token callback (target; event mapping and the approval hand-off are broken today, R-4, R-5, R-24).
- **AI Tutor**: Bedrock retrieval plus a streaming Claude call over Server-Sent Events (history fix applied, R-15; quotas open, R-32). The widget renders with textContent only; no XSS was found.

Review method

CHECK	RESULT
python -m py_compile on all 91 Python files on the review branch	Pass
bash -n on every shell script and the Tutor init task	Pass
Django system check in a scratch harness (plugins linked as chronopoli_*)	Pass. Migrations match the schema; only help_text and label drift remain
URL resolution against the real urls.py files	3 routing defects confirmed (R-2, R-11, R-21)
Terraform reference scan and terraform validate (1.16.4, local provider mirror)	No undefined references; validate passes after the updates, with provider deprecation warnings only
Dependency versions against PyPI, GitHub releases, ghcr.io and Tutor defaults	28 components inventoried (next chapter)
Open edX: Tutor 22.0.2 (Verawood) run locally in Docker with the eight demo courses	All 'Real Open edX screen' figures were captured from this instance
Chronopoli templates rendered by their real views in a Django test harness	Onboarding, academy, jobs, badge and symposia figures; the harness renders Mako templates with a Mako engine (R-9), stubs the Open edX grades API and enables the subdomain middleware (R-23)
Discourse 3.5 and Opencast 20.4 running in Docker	Community and video figures, with district categories as defined in setup-discourse-categories.sh
Design prototypes for pages without a template	Checkout success, team licences, partner dashboard and Digital Twin (R-10); AI Tutor widget on a composed lesson page

CHAPTER 15

Third-party stack and versions

The main third-party components Chronopoli uses, what each one does for the platform, the version pinned before the review and the target version for September 2026.

● URGENT

Three items were urgent. **(1)** The Claude model ID used in 8 places is retired. **(2)** RDS MySQL 8.0 left standard support on 31 July 2026 and now incurs Extended Support fees. **(3)** The plugin's Tutor cap of <22 would have downgraded the platform to an end-of-life Open edX release.

The table lists 28 components. Rows marked 'Tutor default' are the images Tutor 22.0.2 ships with. The Titan Embeddings row (the first row on the next page) is about region availability rather than versions: the model is not offered in me-central-1.

COMPONENT	ROLE IN CHRONOPOLI	PINNED BEFORE	TARGET (SEP 2026)	LICENCE
Tutor	Deploys Open edX with Docker	<22 (plugin)	22.0.2 (Verawood)	AGPL-3.0
Open edX platform	LMS, Studio, MFEs	Follows Tutor; docs said Redwood	release/verawood	AGPL-3.0
Django / Celery	Plugin framework, async tasks	From the Open edX image	Django 5.2 LTS, Celery 5.6	BSD-3
MongoDB	Course content store (Tutor)	Tutor default	7.0.39 (Tutor 22 default)	SSPL-1.0
Redis	Cache and Celery broker (Tutor)	Tutor default	7.4.10 (Tutor 22 default)	RSALv2 / SSPLv1
Meilisearch	Course search (Tutor)	Tutor default	v1.36.0 (Tutor 22 default)	MIT
Caddy	Web server and TLS (Tutor)	Tutor default	2.11.4 (Tutor 22 default)	Apache-2.0
RDS MySQL	Open edX database	8.0	8.4 LTS	GPL-2.0
Terraform CLI	Infrastructure as code	>= 1.5.0	1.16.4	BUSL-1.1
AWS provider	All AWS resources	~> 5.0 (lock 5.100.0)	5.x kept (6.66.0 in sprint 6)	MPL-2.0
archive provider	Zips Lambda code	undeclared	2.8.1	MPL-2.0
Ubuntu (EC2 AMI)	Host OS	22.04 jammy	24.04 LTS noble (26.04 LTS deferred)	various
Docker + Compose	Runs Tutor and extensions	unpinned, two install paths	Ubuntu docker.io + compose-v2	Apache-2.0
AWS Lambda runtime	8 pipeline functions	python3.11	python3.13 (3.14 available)	-
Claude on Bedrock	AI Tutor, analysis, content generation	claude-3-5-sonnet (retired)	Sonnet 4.6 global; Sonnet 5.5 in sprint 6	commercial

COMPONENT	ROLE IN CHRONOPOLI	PINNED BEFORE	TARGET (SEP 2026)	LICENCE
Titan Embeddings v2	Knowledge-base vectors	region: me-central-1	region: eu-central-1 or eu-west-1	commercial
OpenSearch Serverless	Knowledge-base vector index	Terraform, no index	Managed service (no version)	AWS service
boto3	AWS SDK	unpinned	1.43.104	Apache-2.0
stripe	Checkout + Connect	not installed	15.6.1	MIT
WeasyPrint / Jinja2	Opinion-paper PDF	unpinned	70.0 / 3.1.6	BSD-3
ElevenLabs API	Twin voice cloning	REST v1	REST v1	commercial
HeyGen API	Twin avatar video	REST v2	REST v2	commercial
Opencast allinone	Video platform	latest	20.4	ECL-2.0
Presenton	AI slide builder	latest	v0.9.11-beta	Apache-2.0
Discourse	Community forum	tests-passed branch	latest branch (3.5 tested)	GPL-2.0
actions/checkout	CI checkout	v4 (Node 20 runtime, retired)	v7.0.1	MIT
nginx	Proxy for extensions	Ubuntu package	Ubuntu package	BSD-2
Inter / JetBrains Mono	Theme typography	Google Fonts	-	OFL-1.1

CHAPTER 16

Review findings

37 verified findings, sorted by severity. Each one was confirmed by reading the code or by running a check. The 7 findings marked FIXED, the partial fixes and the version updates in chapter 17 were tested and pushed to the review branch in 15 commits.

Summary of findings by severity					
6	CRITICAL		18	HIGH	
	Block go-live			Fix before launch	
			13	MEDIUM	
				Fix around launch	
			7	FIXED NOW	
				On the review branch	

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-1	CRITICAL	Packaging	The eight Django apps in plugins/ are never pip-installed. Folder names (ai-onboarding) do not match module names (chronopoli_onboarding), so LMS/CMS fail with ModuleNotFoundError.	Give each app a pyproject.toml and install it through Tutor's extra pip requirements setting or a Dockerfile patch, then rebuild the image.	Open
R-2	CRITICAL	Routing	URLs are added through a Tutor patch, openedx-lms-urls, that Tutor never renders, so no Chronopoli endpoint is routed.	Register each app as an edx-platform plugin: an lms.djangoapp entry point and a plugin_app dict with url_config and settings_config.	Open
R-3	CRITICAL	Pipeline	The CheckTranscription ResultSelector expects transcript keys that the Lambda returns only after COMPLETED. The first poll fails with States.Runtime.	Always return both keys, as empty strings until the job completes.	FIXED
R-4	CRITICAL	Pipeline	The EventBridge input reads fields that the IVS event does not send. The recording prefix points at HLS segments, not an MP4, and round_table_id becomes the channel name.	Map the bucket-name and key-prefix fields of the IVS event, render an MP4 (IVS rendition or MediaConvert), and carry the RoundTable ID in a channel tag.	Open
R-5	CRITICAL	Pipeline	The approval hand-off is broken end to end: no waitForTaskToken, a dict vs list schema mismatch, no task_token stored, and the poller is never scheduled.	Use the waitForTaskToken integration of lambda:invoke, agree one message schema, store the token and call send_approval from the approve view.	Open

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-6	CRITICAL	Security	The media bucket is public-read and also holds grade CSVs (learner PII), course exports and expert documents.	Keep media private behind CloudFront OAC with signed URLs, and move grades and exports to a private bucket.	Open
R-7	HIGH	Settings	getattr(globals(), ...) always returns {}, which wipes the Celery beat schedule. The referenced weekly-report task does not exist.	Fixed with globals().get() and a new send_all_weekly_reports task. Tutor still needs a beat process.	FIXED
R-8	HIGH	Tutor	The init task is never registered, and several scripts call manage.py without the lms argument. backup.sh always aborts.	Register it with CLI_DO_INIT_TASKS and use ./manage.py lms ...	FIXED
R-9	HIGH	Templates	Mako templates are rendered by the Django engine and show up as literal text. \${csrf_token} breaks onboarding CSRF.	Render them with edxmako render_to_response; make the Django templates standalone.	Open
R-10	HIGH	Templates	Templates are missing for ecommerce success/cancel/subscriptions, the partner dashboard and all masterclass pages, so those pages return 500.	Create the templates.	Open
R-11	HIGH	E-commerce	checkout/<course_key>/ is matched before checkout/success/, so Stripe's success_url returns 404.	Order the routes so success/ and cancel/ come first.	FIXED
R-12	HIGH	E-commerce	Payment is not enforced inside Open edX. Learners can enrol for free through the standard LMS button or API.	Add an openedx-filters enrolment step that rejects unpaid enrolments in verified-only courses; removing the audit mode alone is not enough.	Open
R-13	HIGH	E-commerce	stripe_transfer_id is unique with a default of '', so only the first partner payout is ever saved. Status is never set to paid.	Make it nullable, store None, and update the status from Stripe webhooks.	PARTIAL
R-14	HIGH	E-commerce	The webhook marks purchases paid and enrolled without checking payment_status, and even when enrolment failed. Refunds are ignored.	Check payment_status == paid, return the real enrolment result, and handle async payments and refunds.	PARTIAL
R-15	HIGH	AI Tutor	The history window can start with an assistant turn, which Bedrock rejects. The AI Tutor fails from about the sixth question onwards.	Trim the window to start at a user turn and merge consecutive turns from the same role.	FIXED

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-16	HIGH	IAM	bedrock-agent:Retrieve is not an IAM action, so knowledge-base retrieval is denied. StartIngestionJob is missing.	Grant Retrieve, RetrieveAndGenerate and StartIngestionJob under the bedrock prefix.	FIXED
R-17	HIGH	IAM	The EC2 role lacks SES, SQS, IVS, Step Functions and S3 (knowledge/outputs) permissions that the Django apps need.	Add a least-privilege policy for each service.	Open
R-18	HIGH	Terraform	The Bedrock knowledge base has no OpenSearch data-access policy and no vector index, so apply fails. The network policy is public.	Add the access policy and create the index; restrict network access.	Open
R-19	HIGH	Terraform	The CloudFront origin is an IP address, which is not allowed. An http-only origin combined with Caddy's HTTPS redirect would cause a redirect loop.	Use a DNS origin (origin.learn...) with https-only.	Open
R-20	HIGH	Lambda	The opinion-paper Lambda imports jinja2 and weasyprint, but only handler.py is zipped, and WeasyPrint needs native libraries.	Ship it as a container-image Lambda (or a layer).	Open
R-21	HIGH	Symposia	The approval UI called a URL that did not resolve, and the routes are prefixed twice (/chronopoli/symposia/symposia/...).	Use {% url %} (done, e4072ed); drop the doubled prefix in symposia/urls.py (open).	PARTIAL
R-22	HIGH	Academy	The auto-badge signal imports CERTIFICATE_CREATED from a module that does not export it, so auto-badging is silently disabled.	Use openedx_events CERTIFICATE_CREATED and read certificate.user / course_key.	Open
R-23	HIGH	Academy	The subdomain middleware is never registered, matches slug rather than subdomain, and has no Caddy host, TLS or ALLOWED_HOSTS entry.	Register it, match on subdomain + academy_enabled, and add wildcard hosting.	Open
R-24	HIGH	Symposia	IVS channels are created without a recording configuration. create_channel and start_pipeline are never called, and the input keys do not match.	Attach an IVS recording configuration (by ARN), wire the calls and use unique execution names.	Open
R-25	MEDIUM	Privacy	The intelligence-reports API returns every partner's reports to any partner.	Filter by partner.	PARTIAL

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-26	MEDIUM	Auth	Dashboard access is granted when the user's e-mail matches the partner contact e-mail, even though that address is never verified.	Add an explicit partner-user relation.	Open
R-27	MEDIUM	Auth	Discourse SSO forwards unverified e-mails, which allows a takeover of an existing forum account.	Send require_activation for inactive accounts.	FIXED
R-28	MEDIUM	Config	The Discourse SSO secret is generated by Tutor, but setup-discourse.sh reads it from an environment variable. Nothing keeps the two in sync.	Read it with tutor config printvalue.	Open
R-29	MEDIUM	Privacy	Draft round tables and unapproved AI outputs are visible to any logged-in user.	Restrict them to staff and the initiator.	Open
R-30	MEDIUM	Masterclass	Unpublished twins are visible, and the upload view is a stub. Synchronous Textract fails on multi-page PDFs.	Add a publish check, a real upload and async Textract.	Open
R-31	MEDIUM	Settings	AWS_REGION, S3_MEDIA_BUCKET, SYMPOSIA_SFN_ARN and similar settings are read but never defined, and region defaults disagree.	Define them in the plugin settings patch.	Open
R-32	MEDIUM	AI Tutor	Bad JSON gives a 500. course_key goes unvalidated into the system prompt, and there is no rate limit on paid Bedrock calls.	Validate input and add a per-user quota.	Open
R-33	MEDIUM	Onboarding	Posted answers are not checked against the allowed options.	Validate each answer against the allowed options in ONBOARDING_QUESTIONS before saving.	Open
R-34	MEDIUM	Scripts	configure-tutor.sh can block on stdin, overwrites a tracked patch file and sets public-read ACLs.	Refactor the script alongside R-6.	Open
R-35	MEDIUM	Terraform	SSM parameters default to empty strings, which SSM rejects.	Create them conditionally or make the variables required.	Open
R-36	MEDIUM	CI/CD	The deploy workflows restart Tutor but never run migrations.	Run tutor local do init --limit=chronopoli.	Open
R-37	MEDIUM	Privacy	Badge assertion IDs are inconsistent and recipient hashes are unsalted. The talent pipeline auto-applies learners without consent (PDPL risk).	Fix the assertion data and add an explicit opt-in.	Open

CHAPTER 17

Updates applied and road to go-live

What changed on the review branch, and a six-sprint plan that covers every open and partial finding.

Updates applied on the review branch

AREA	BEFORE	AFTER
Tutor / Open edX	tutor>=18,<22 (Ulmo, EOL)	tutor>=22.0,<23 (Verawood); Python>=3.10
Tutor install	unpinned tutor[full], system pip	tutor[full]>=22.0.2,<23 in /opt/chronopoli/venv
Terraform	>= 1.5.0; archive provider undeclared	>= 1.14.0; archive ~> 2.8 locked (validate passes on 1.16.4)
EC2 AMI	Ubuntu 22.04 jammy	Ubuntu 24.04 noble (gp3)
Docker packages	docker-compose-plugin (not in Ubuntu)	docker-compose-v2 + python3-venv
RDS MySQL	8.0 (extended support fees)	8.4 LTS
Lambda runtime	python3.11 (8 functions)	python3.13 (8 functions)
Lambda deps	unpinned; unused anthropic package	boto3 1.43.104, weasyprint 70.0, jinja2 3.1.6
Claude on Bedrock	claude-3-5-sonnet-20241022 (retired), hardcoded in 8 places	global.anthropic.claude-sonnet-4-6 via BEDROCK_MODEL_ID
Bedrock IAM	bedrock-agent:Retrieve (invalid)	bedrock:Retrieve / RetrieveAndGenerate / StartIngestionJob + global profiles
Opencast	allinone:latest, DB_VENDOR MySQL	allinone:20.4, MariaDB driver
Presenton	presenton:latest	presenton:v0.9.11-beta
Discourse	tests-passed branch	latest (renamed channel)
GitHub Actions	actions/checkout@v4 (Node 20)	actions/checkout@v7
Stripe SDK	not installed	stripe==15.6.1 via OPENEDX_EXTRA_PIP_REQUIREMENTS; webhook adapted to 15.x objects
Docs	Redwood, PostgreSQL 14	Verawood (Tutor 22), MySQL 8.4

The review branch is `claude/zealous-knuth-y3ddq4` in the Chronopoli repository.

● HELD BACK

Deliberately **not** changed yet: AWS provider 5.x to 6.x (a breaking major release that needs a reviewed terraform plan) and Claude Sonnet 5.5 (needs the temperature parameter removed from the Lambdas and a regional availability check). Both are planned for sprint 6.

Six-sprint plan to go-live

SPRINT	GOAL	FINDINGS
1 Boot	Package the Django apps as edx-platform plugins; fix routing, templates, settings, the badge signal and answer validation; add the Celery beat service to Tutor (R-7)	R-1, R-2, R-9, R-10, R-21, R-22, R-31, R-33
2 Secure	Private media and exports, IAM least privilege, partner access model, SSO secret, script ACLs, consent	R-6, R-17, R-25, R-26, R-28, R-29, R-34, R-37
3 Commerce	Paid course modes, payout status from Stripe webhooks, refunds and webhook hardening	R-12, R-13, R-14
4 Pipeline	IVS recording, event mapping, task tokens, container Lambda for PDFs	R-4, R-5, R-20, R-24
5 AI & data	Knowledge-base index and access policy, twin publishing and uploads, AI Tutor quotas; move Titan embeddings to an EU region (see chapter 15)	R-18, R-30, R-32
6 Launch	CloudFront DNS origin, wildcard academies, SSM defaults, CI migrations; AWS provider 6.x and the Claude Sonnet 5.5 migration	R-19, R-23, R-35, R-36

The plan covers all 30 open or partial findings, plus the Celery beat service that R-7 still needs; the other 6 fixed findings need no further work.

— CHAPTER 18

Glossary

Short definitions of the terms used in this booklet.

TERM	MEANING
Bedrock	Amazon's managed service for foundation models, used here to run Claude
Company Academy	A partner's white-label space with pathways, badges and job postings on its own subdomain
Course mode	An Open edX enrolment track of a course run (for example honor or verified) with its price
CSCS	Certified Stablecoin Compliance Specialist, the badge of the illustrative Tether compliance pathway
Digital Twin	An expert-approved AI version of a person, trained on their published work
District	An Open edX organisation (for example CHRON-COMP) that groups courses by field of knowledge
FIU	Financial Intelligence Unit, the national agency that receives suspicious transaction reports
GST	Gulf Standard Time, UTC+4
IVMS101	The interVASP messaging standard that defines Travel Rule data fields
Layer	Depth of a course, from L1 Entry to L4 Influence
MFE	Micro-frontend: a separate Open edX web app such as the Learning app, Studio authoring or the catalogue
MiCA	The EU Markets in Crypto-Assets Regulation
MLRO	Money Laundering Reporting Officer, responsible for a firm's AML reporting
Open Badge	A verifiable digital credential following the 1EdTech (formerly IMS Global) Open Badges 2.0 standard
Pathway	An ordered set of courses that ends with a badge
PDPL	The UAE Personal Data Protection Law (Federal Decree-Law No. 45 of 2021)
Round table / Symposium	A closed expert session under the Chatham House Rule, streamed to registered participants, with AI-generated outputs
Travel Rule	FATF Recommendation 16: originator and beneficiary data must travel with a transfer
Tutor	The official Docker-based distribution and deployment tool for Open edX
USDT	Tether's US-dollar stablecoin, used here only as an illustrative example
VARA	Dubai's Virtual Assets Regulatory Authority
VASP	Virtual Asset Service Provider, such as an exchange or custodian